3.4 Financial Time Series

Concepts of Time Series

Time Series Data

- A time series $\{Y_t\}$ is a continuous state, discrete time process where t = 1,2 … , T
- In Financial time series Y can be Stock price, Interest Rates, FX rate, commodity prices etc.

Time Series Data

- We want to fit statistical models to predict future probability distribution and this is needed for three reasons
  - Derivative Pricing (e.g. Option Pricing)
  - Risk Management (Especially the future variance is important for risk management)
  - Trading

Time Series Data

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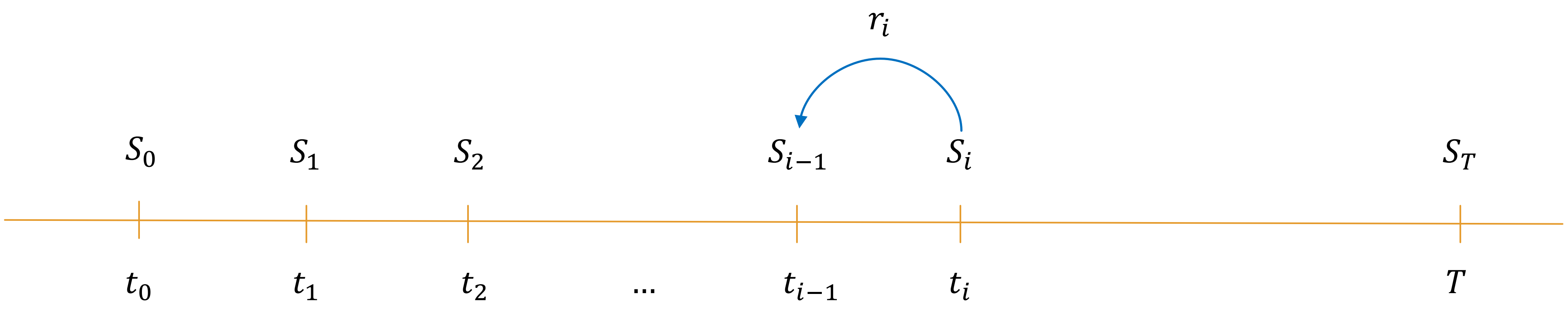
Arithmetic returns vs Geometric Returns

- Consecutive prices are correlated and variance of prices normally increase with time.
- That's why we focus on returns or price changes from a modeling standpoint.



Daily Arihmetic return $$r_i = \frac{S_i - S_{i-1}}{S_{i-1}} = \frac{S_i}{S_{i-1}} - 1$$

T – days Arihmetic return $$R = \frac{S_T}{S_0} - 1 = \frac{S_T}{S_{T-1}} \frac{S_{T-1}}{S_{T-2}} \ldots \frac{S_1}{S_0} - 1 = (1 + r_1)(1 + r_2) \ldots (1 + r_T) - 1$$

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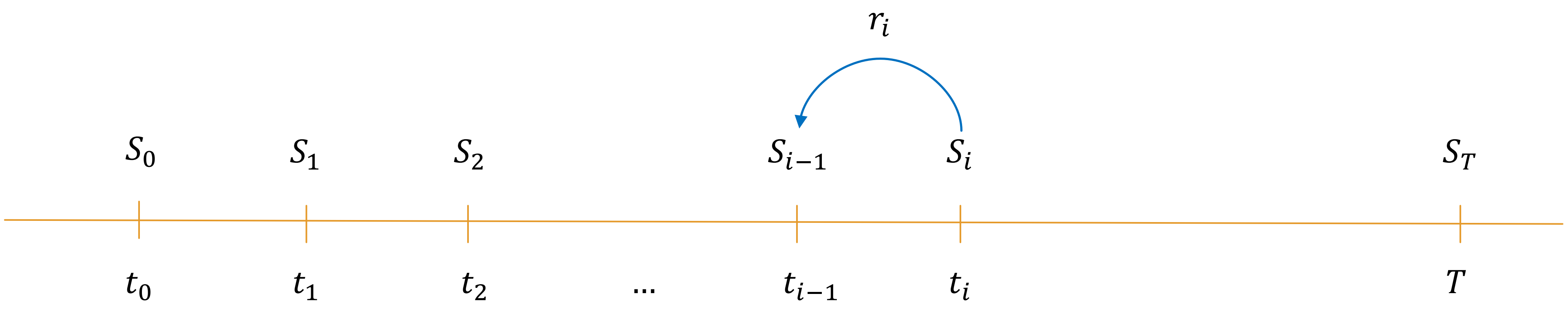

Daily Arihmetic return $\quad r_i = \dfrac{S_i - S_{i-1}}{S_{i-1}} = \dfrac{S_i}{S_{i-1}} - 1$

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$$S_T \to 0, R \to -100\%$$

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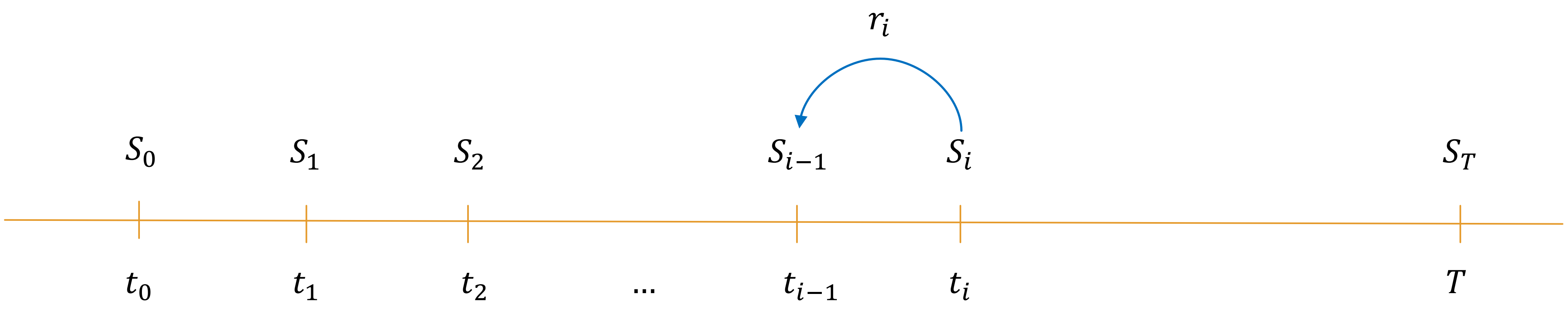


Daily Geometric return $\quad d_i = \ln\left(\frac{S_i}{S_{i-1}}\right)$

T – days Geometric return $\quad D = \ln\left(\frac{S_T}{S_0}\right) = \ln\left(\frac{S_T}{S_{T-1}}\frac{S_{T-1}}{S_{T-2}}\dots\frac{S_1}{S_0}\right) = \ln\left(\frac{S_T}{S_{T-1}}\right) + \ln\left(\frac{S_{T-1}}{S_{T-2}}\right) + \cdots + \ln\left(\frac{S_1}{S_0}\right)$

$$= d_1 + d_2 + \cdots + d_T$$

$$S_T \to 0, D \to -\infty$$

Arithmetic returns vs Geometric Returns

- When time resolution is high (time steps are smaller) and price volatility is less, then consecutive price changes are small.
- In this case, the arithmetic and the geometric returns will be similar.

Daily Geometric return $d_i = \ln\left(\frac{S_i}{S_{i-1}}\right)$

Daily Arihmetic return $r_i = \frac{S_i}{S_{i-1}} - 1$

$$d_i = \ln(1 + r_i)$$

Taylor Series

$$\ln(1 + x) = x - \frac{1}{2}x^2 + \cdots$$

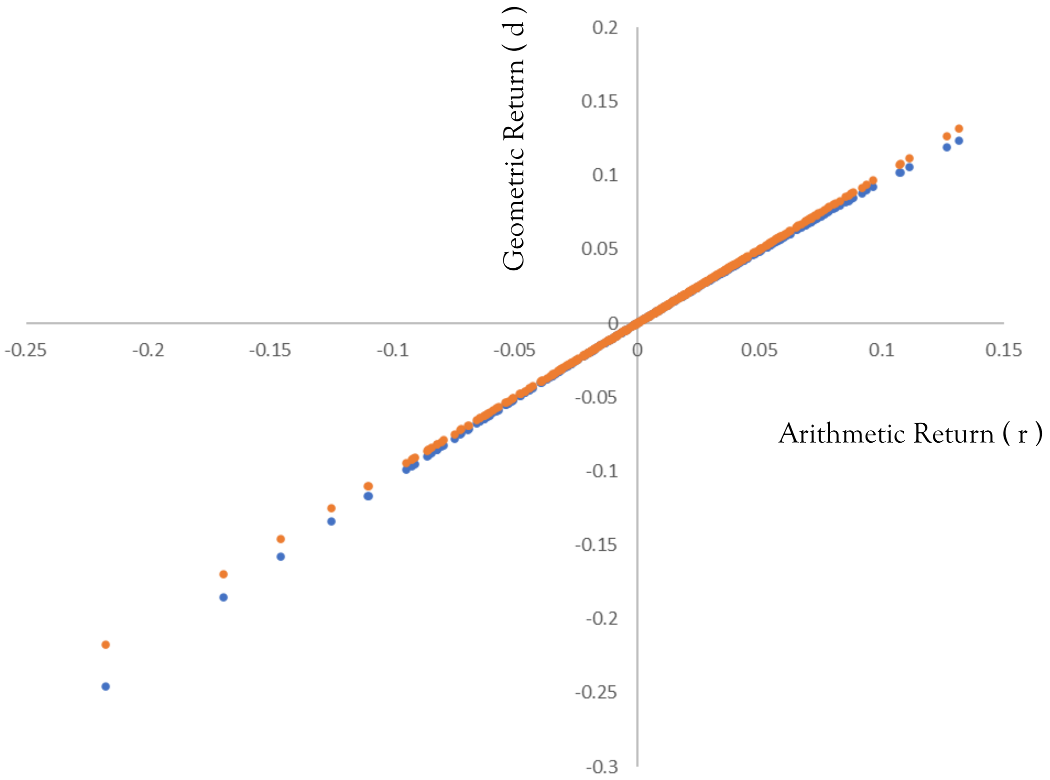
$$\approx x$$

$$d_i \approx r_i$$

Arithmetic returns vs Geometric Returns

- When time resolution is high (time steps are smaller) and price volatility is less, then consecutive price changes are small.
- In this case, the arithmetic and the geometric returns will be similar.

Date	Adj. Close	r	d
1-Mar-22	4,530.41	0.035773	0.035148
1-Feb-22	4,373.94	-0.03136	-0.03186
1-Jan-22	4,515.55	-0.05259	-0.05402
1-Dec-21	4,766.18	0.043613	0.042689
1-Nov-21	4,567.00	-0.00833	-0.00837
1-Oct-21	4,605.38	0.069144	0.066858
1-Sep-21	4,307.54	-0.04757	-0.04874
…			
1-Jun-85	191.85	0.012134	0.012061
1-May-85	189.55	0.054051	0.052641
1-Apr-85	179.83	-0.00459	-0.0046
1-Mar-85	180.66	-0.00287	-0.00287
1-Feb-85	181.18	0.008629	0.008592

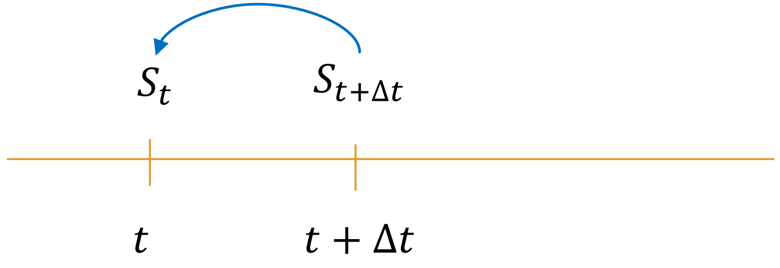


Modeling Geometric Returns

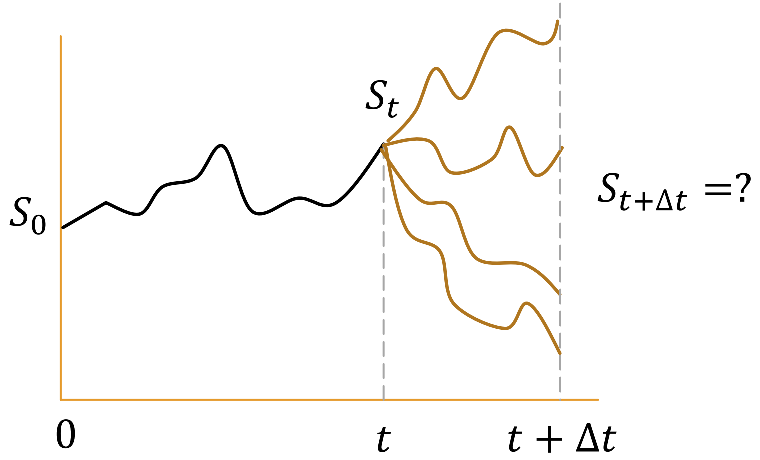
- It's common to model geometric returns as normally distributed. (One of the assumptions of Black Scholes Option Pricing)
- If geometric returns are normal, prices and arithmetic returns will be log normally distributed.

Black Scholes World

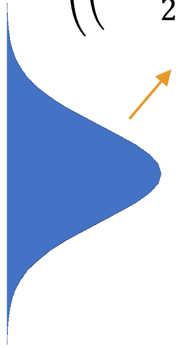
$$\ln\left(\frac{S_{t+\Delta t}}{S_t}\right)$$



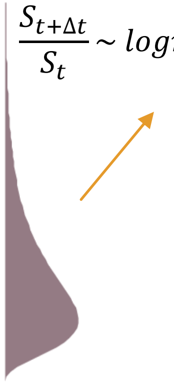
$$\ln\left(\frac{S_{t+\Delta t}}{S_t}\right) \sim \mathcal{N}\left(\underbrace{\left(r-\frac{1}{2}\sigma^2\right)\Delta t}_{\text{mean}},\ \underbrace{\sigma^2\Delta t}_{\text{variance}}\right)$$



$$\ln\left(\frac{S_{t+\Delta t}}{S_t}\right) \sim \mathcal{N}\left(\left(r-\frac{1}{2}\sigma^2\right)\Delta t,\quad \sigma^2\Delta t\right)$$



$$\frac{S_{t+\Delta t}}{S_t} \sim lognormal\left(\left(r-\frac{1}{2}\sigma^2\right)\Delta t,\quad \sigma^2\Delta t\right)$$



Modeling Arithmetic Returns

- In Markowitz world, (commonly denoted as Mean-variance utility framework) portfolio returns and variance are based off of arithmetic returns and not geometric returns. Also, the portfolio risk is assumed to be completely described by the variance which is possible only if returns are assumed multivariate normal.

Markowitz World

Portfolio

Stock	Weight
S_1	W_1
S_2	W_2
…	…
S_n	W_n

Portfolio Return : $r_p = \boldsymbol{W}^T \boldsymbol{r}$

$$if\ \boldsymbol{r} \sim \mathcal{N}\ (\boldsymbol{\mu}\ , \boldsymbol{\Sigma})$$

$$then\ \ r_p \sim \mathcal{N}\ \left(\boldsymbol{W}^{\boldsymbol{T}}\boldsymbol{\mu}\ , \boldsymbol{W}^T \boldsymbol{\Sigma} \boldsymbol{W}\right)$$

Portfolio Mean

Portfolio Variance

Mean variance utility

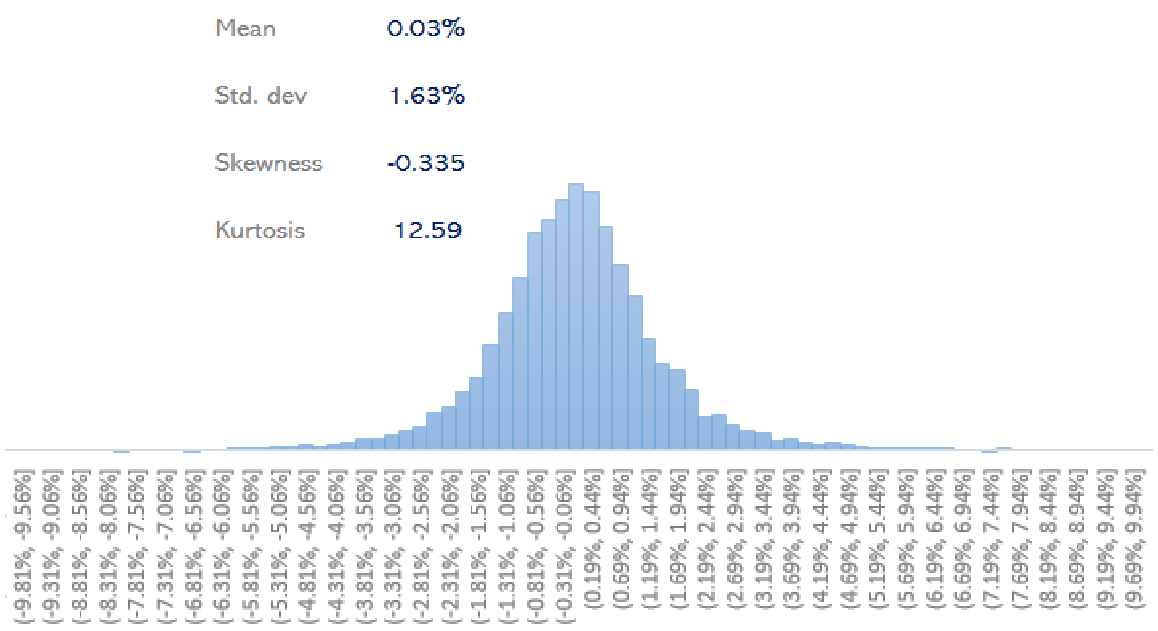
risk aversion

$$\max_{\boldsymbol{W}} \left(\boldsymbol{W}^{\boldsymbol{T}}\boldsymbol{\mu} - \lambda \boldsymbol{W}^T \boldsymbol{\Sigma} \boldsymbol{W}\right)$$

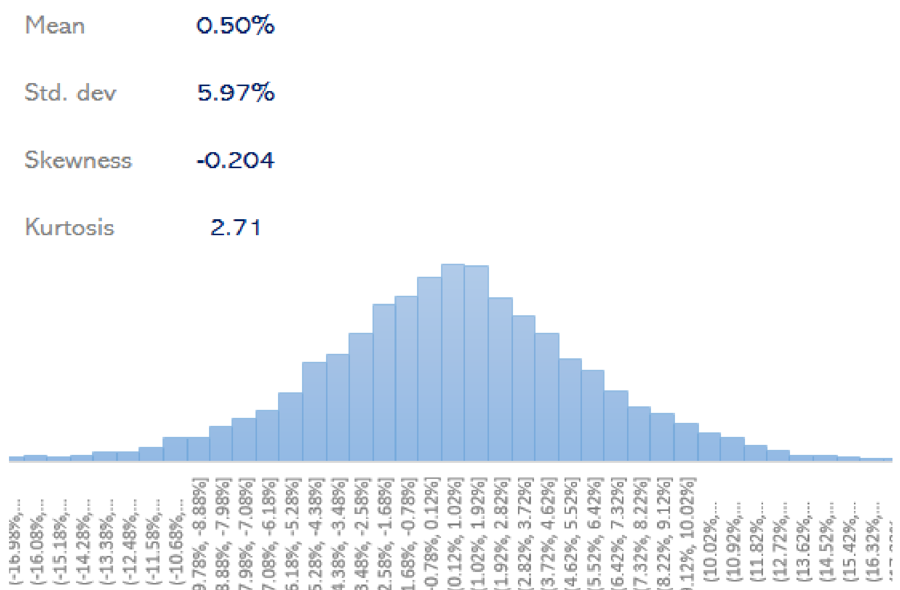
$$s.t. \sum W_i = 1$$

Aspects of Time

- Financial time Series models are affected by both time resolution and time horizon.
- Time resolution refers to how densely data is collected which can vary from second by second or year by year.
- For short time horizons (e.g. intraday, daily or weekly) , returns show heavier tail which becomes important for Market risk (e.g. Value at Risk) perspective. If one assumes normal distribution for short term returns, it will underestimate VaR.
- As we go from daily returns to monthly returns, distribution becomes more and more normal.



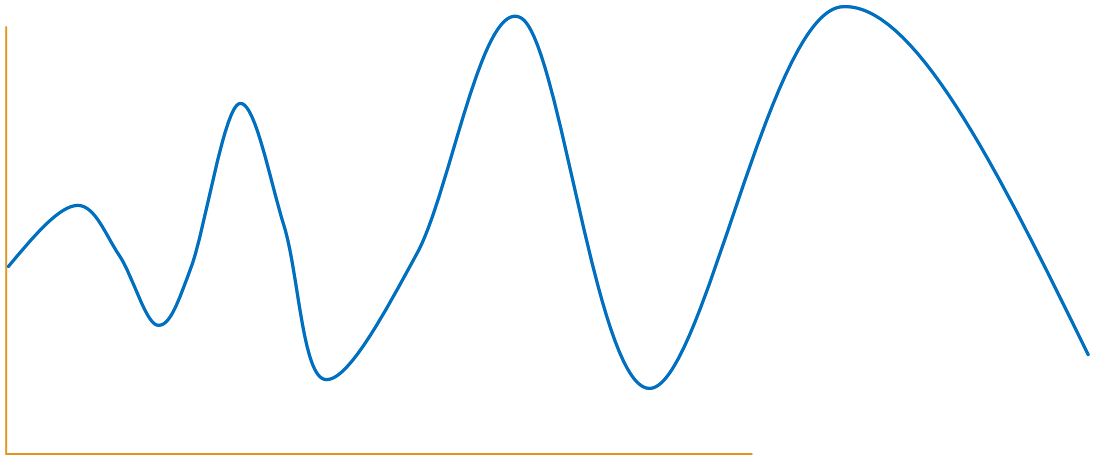
IBM Daily returns



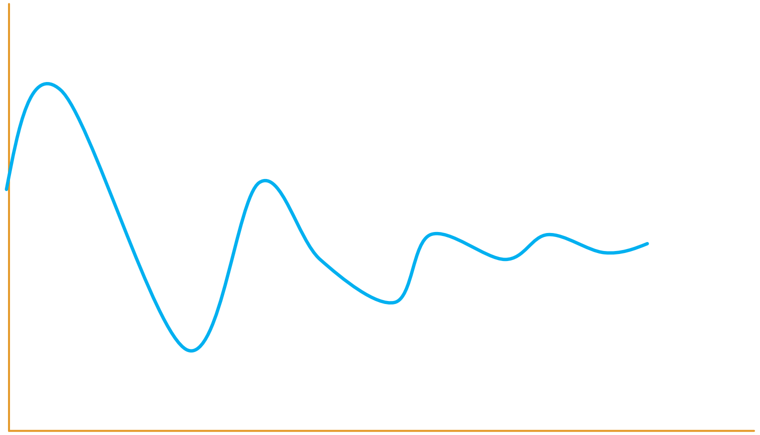
IBM Bi – Weekly returns

AR Models

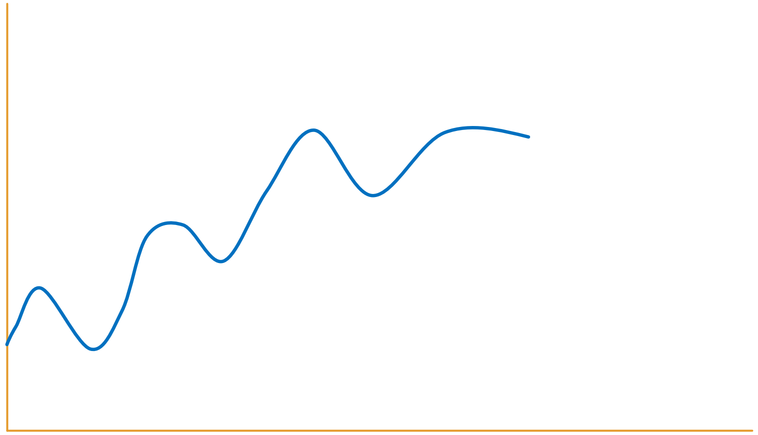
Building Blocks of Time Series



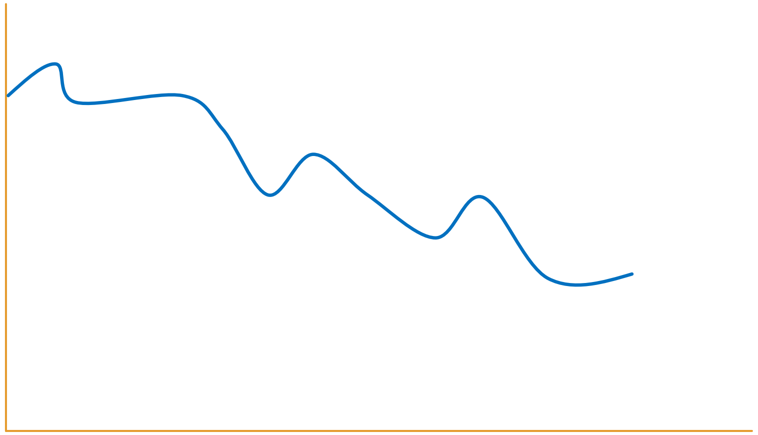
Increasing Variance



Diminishing Variance



Trending Up



Trending Down

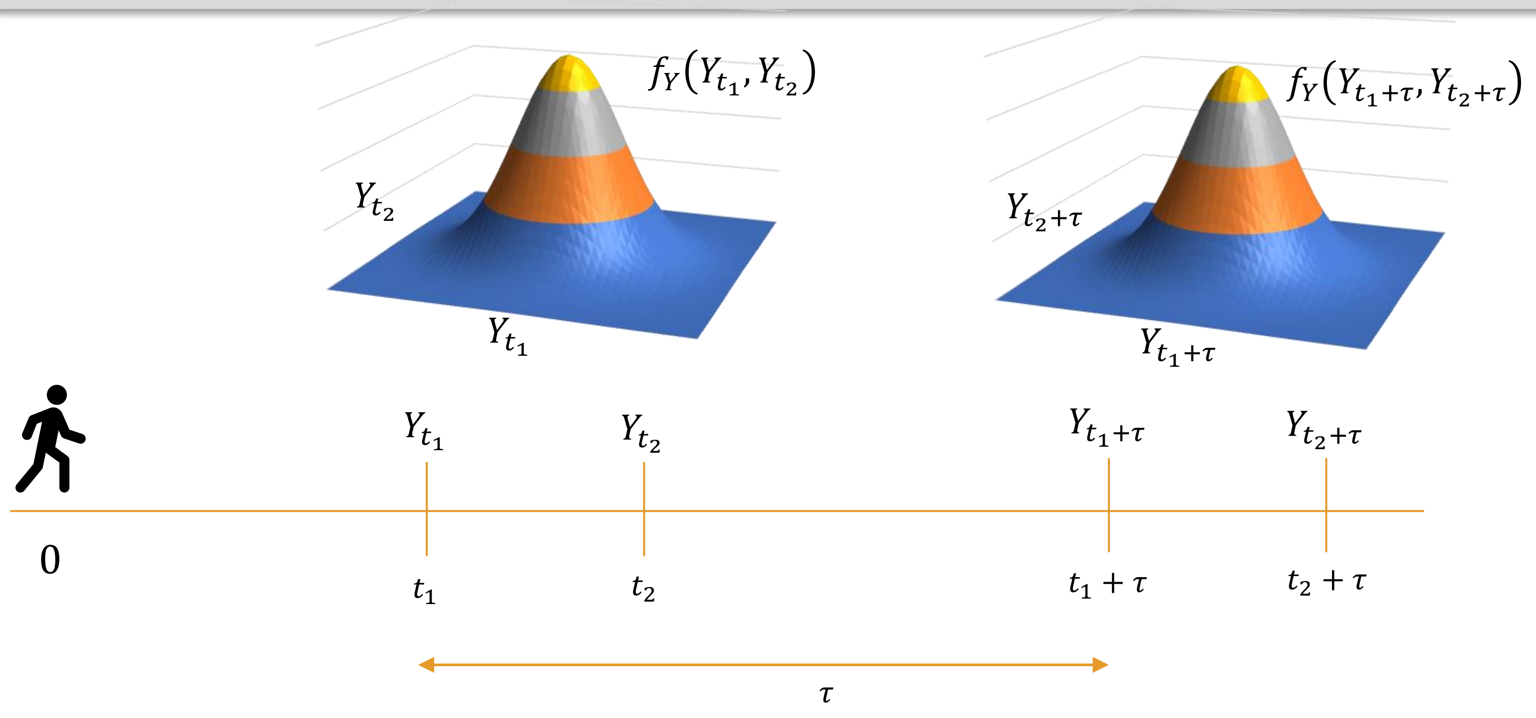
Building Blocks of Time Series

- Stationarity

Strong Stationarity / Strict Stationarity

If $\{Y_t\}$ is a time series, then the unconditional joint distribution of $(Y_{t_1+\tau}, Y_{t_2+\tau}, \dots, Y_{t_n+\tau})$ is same as $(Y_{t_1}, Y_{t_2}, \dots, Y_{t_n})$ which means that the distribution does not change as time passes

$$f_Y(Y_{t_1+\tau}, Y_{t_2+\tau}, \dots, Y_{t_n+\tau}) = f_Y(Y_{t_1}, Y_{t_2}, \dots, Y_{t_n})$$



Building Blocks of Time Series

- Stationarity

Weak Stationarity / Covariance Stationarity

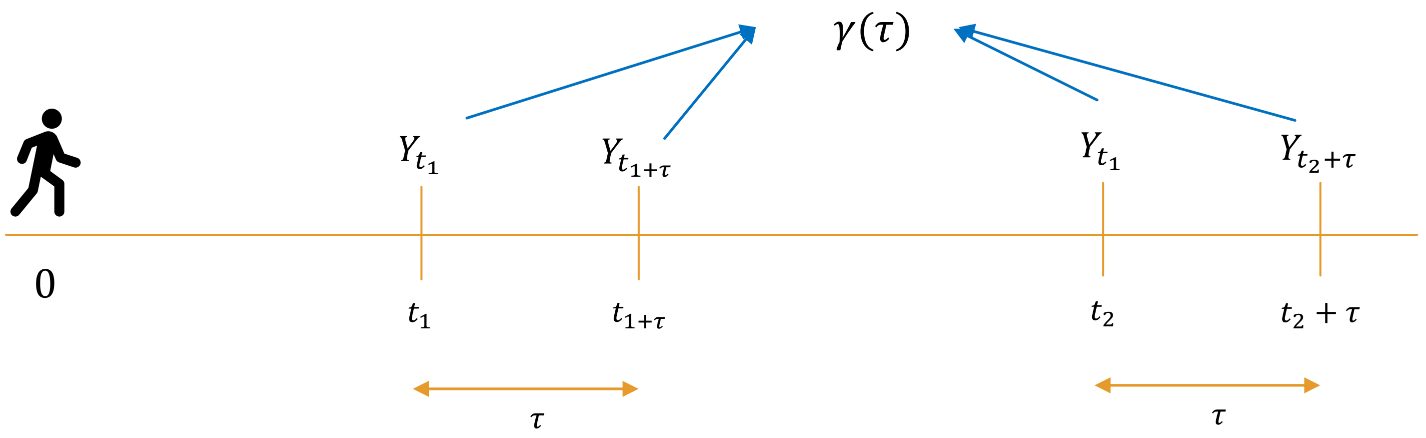
If $\{Y_t\}$ is a time series, then weak stationarity refers to the property where the unconditional mean is constant and the autocovariance only depends on the time lag between the returns no matter where they are in future.

$\mathbb{E}[Y_t] = \mathbb{E}[Y_{t+\tau}] = \mu$ and

$Cov\left(Y_{t_1}, Y_{t_1+\tau}\right)$ depends only on τ e.g. $Cov\,(r_1, r_2) = Cov\,(r_{99}, r_{100}) = \gamma(1)\ \ or\ \ \gamma_1$

$\gamma_0 = Cov\,(r_t, r_t) = Var(r_t)$

AutoCovariance



Building Blocks of Time Series

- White Noise Process

- A time series $\{Y_t\}$ is called a White Noise process if it's a sequence of *iid* random variables.
- Specifically if it's iid normal, we call it Gaussian White Noise process.
- White Noise process follows strong stationarity in a trivial way (All Means and Covariances are 0 , hence constant)

- Linear Time Series

- A linear time series $\{Y_t\}$ is one which can be represented as

$r_t = \phi_0 + \phi_1\epsilon_1 + \phi_2\epsilon_2 + \cdots$ where $\epsilon_i \sim \mathcal{N}\ (0\ , \sigma^2)$

Building Blocks of Time Series

- Auto correlation function (ρ_l)
- Auto correlation function (ACF) for lag l is given as $\rho_l = \frac{\gamma_l}{\gamma_0}$. (Note : $\gamma_l = \gamma_{-l}$ also $\rho_l = \rho_{-l}$)

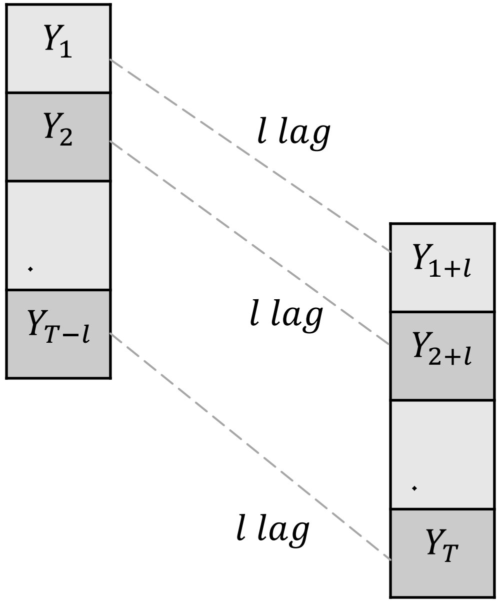
t	Y
1	Y_1
2	Y_2
.	.
T	Y_T

$$\bar{Y} = \sum_{i=1}^{i=T} Y_i$$

Mean estimate

$$\hat{\gamma}_0 = \frac{1}{T}\sum_{i=1}^{i=T}\left((Y_i - \bar{Y})^2\right)$$

Variance estimate



$$\hat{\gamma}_l = \frac{1}{T}\sum_{t=l+1}^{t=T}(Y_t - \bar{Y})(Y_{t-l} - \bar{Y})$$

Auto Variance estimate

$$\hat{\rho}_l = \frac{\hat{\gamma}_l}{\hat{\gamma}_0} = \frac{\sum_{t=l+1}^{t=T}(Y_t - \bar{Y})(Y_{t-l} - \bar{Y})}{\sum_{i=1}^{i=T}\left((Y_i - \bar{Y})^2\right)}$$

ACF for lag l estimate

Building Blocks of Time Series

- Hypothesis Testing for ACF

For time series analysis, it's important that it should be auto-correlated. If there is no auto correlation for any lag, then it's a white noise process and there is nothing to model

We can test ACF individually or jointly.

Testing Individual ACF

$H_0: \rho_l = 0$

$H_1: \rho_l \neq 0$

$$t - ratio = \frac{\hat{\rho}_l}{\sqrt{\left(1 + 2\sum_{i=1}^{l-1} \hat{\rho}_i^{\,2}\right)/T}} \quad \sim \mathcal{N}\ (0,1)$$

Joint ACF Test (**Portmanteau Test**)

$H_0: \rho_1 = \rho_2 = \cdots = \rho_m = 0$

$H_1: \rho_i \neq 0$

Ljung and Box suggested this statistic

$$Q(m) = T(T+2)\sum_{l=1}^{l=m} \frac{\hat{\rho}_l^{\,2}}{T-l} \quad \sim \chi^2\ (m)$$

The AR(p) Model

- Autoregressive (AR) models

Auto regressive models in time series are the most basic time series models where Y is modeled as a linear function of its lags.

AR (p) Model

$$\underbrace{Y_t = \phi_0}_{\text{Intercept}} + \underbrace{\phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p}}_{\text{p − lags}} + \underbrace{\epsilon_t}_{\text{shock}} \qquad \epsilon_t \sim \mathcal{N}\ (0\,, \sigma^2)$$

AR (1) Model

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t \qquad \epsilon_t \sim \mathcal{N}\ (0\,, \sigma^2)$$

The AR(1) Model

AR (1) Model

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t \qquad \epsilon_t \sim \mathcal{N}\ (0\,,\sigma^2)$$

$$\underset{\mu}{\underset{\downarrow}{\mathbb{E}[Y_t]}} = \phi_0 + \phi_1 \underset{\mu}{\underset{\downarrow}{\mathbb{E}[Y_{t-1}]}} + \underset{0}{\underset{\downarrow}{\mathbb{E}[\epsilon_t]}}$$

$$var[Y_t] = {\phi_1}^2 var[Y_{t-1}] + var[\epsilon_t]$$

$$\mu = \phi_0 + \phi_1\mu + 0$$

$${\sigma_Y}^2 = {\phi_1}^2{\sigma_Y}^2 + \sigma^2$$

$$\boxed{\mu = \frac{\phi_0}{1-\phi_1}}$$

Unconditional Mean

$$\boxed{{\sigma_Y}^2 = \frac{\sigma^2}{\left(1-{\phi_1}^2\right)}}$$

Unconditional Variance

$$\mathbb{E}[Y_t|Y_{t-1}] = \phi_0 + \phi_1 Y_{t-1}$$

Conditional Mean

$$var[Y_t|Y_{t-1}] = \sigma^2$$

Conditional Variance

$$\boxed{|\phi_1| < 1}$$

The AR(1) Model

AR (1) Model

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t \qquad \epsilon_t \sim \mathcal{N}\ (0\,,\sigma^2)$$

$$cov\ (Y_t, Y_{t-l}) = cov\ (\phi_1 Y_{t-1}, Y_{t-l}) + cov\ (\epsilon_t, Y_{t-l})$$

$\gamma_l = \phi_1 \gamma_{l-1}$ recurrence relation $\longrightarrow$ $\gamma_l / \gamma_0 = \phi_1 \gamma_{l-1} / \gamma_0 \qquad \rho_l = \phi_1 \rho_{l-1}$

$$\gamma_l = \phi_1 \gamma_{l-1} = {\phi_1}^2 \gamma_{l-2} = {\phi_1}^3 \gamma_{l-3} = \cdots = {\phi_1}^l \gamma_0$$

$$\rho_l = \phi_1 \rho_{l-1} = {\phi_1}^2 \rho_{l-2} = {\phi_1}^3 \rho_{l-3} = \cdots = {\phi_1}^l \rho_0 = {\phi_1}^l$$

$$\boxed{\gamma_l = \frac{{\phi_1}^l}{1 - {\phi_1}^2}}$$

$$\boxed{\rho_l = {\phi_1}^l}$$

The AR(1) Model – Mean Reversion and Persistence

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t$$ AR (1) Model

$$Y_t = \frac{\phi_0}{1-\phi_1}(1-\phi_1) + \phi_1 Y_{t-1} + \epsilon_t$$

$$Y_t = \mu(1-\phi_1) + \phi_1 Y_{t-1} + \epsilon_t$$

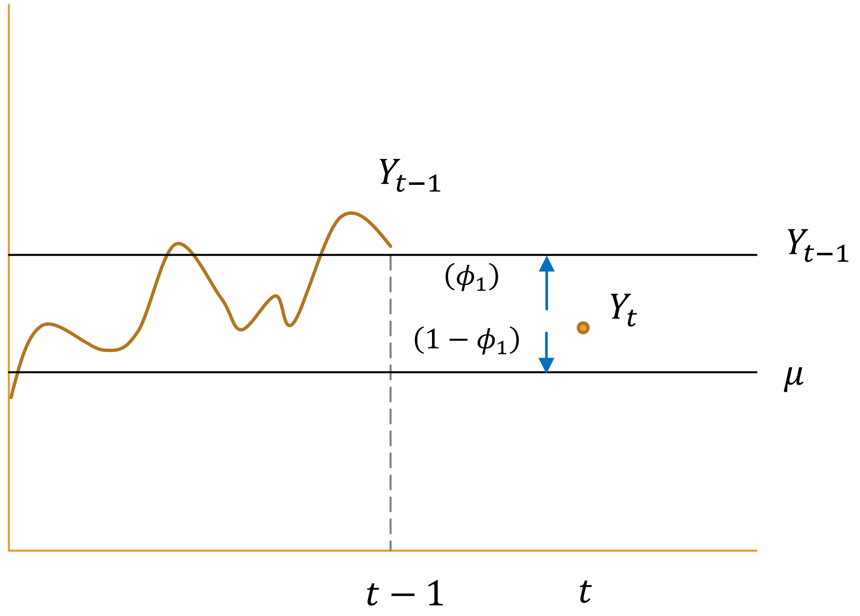
$(1-\phi_1)$ Part listens to unconditional mean μ

(ϕ_1) Part listens to the current level

Shock / Innovation

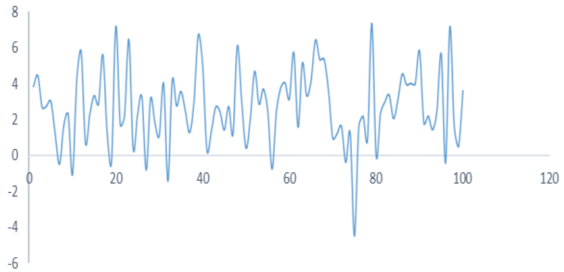
$(\textit{speed of mean reversion})$

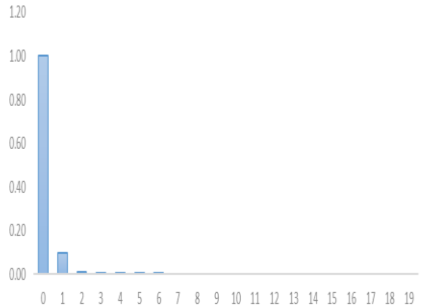
$(\textit{persistence})$



The AR(1) Model

$Y_t = 2 + 0.1Y_{t-1} + \epsilon_t$

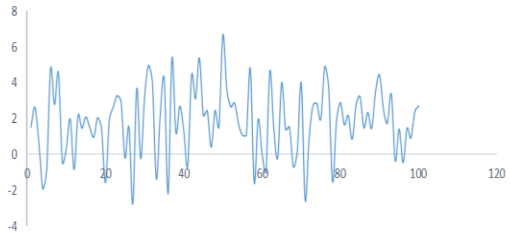


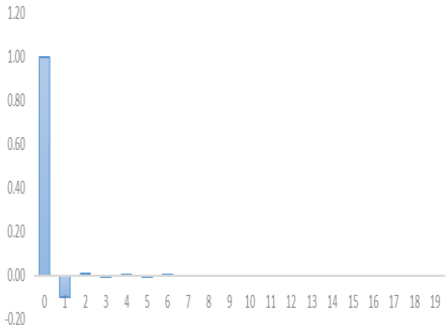


Faster mean reversion (0.9)

Exponentially Decaying ACF

$Y_t = 2 - 0.1Y_{t-1} + \epsilon_t$

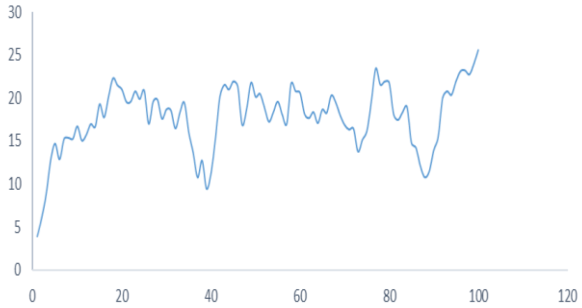


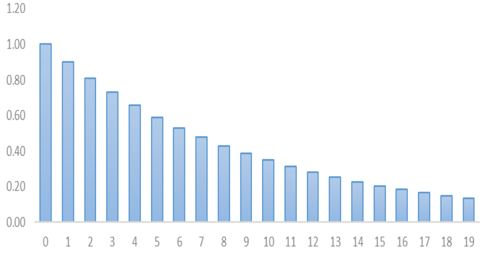


Faster mean reversion (1.1)

Alternate +ve / -ve

$Y_t = 2 + 0.9Y_{t-1} + \epsilon_t$

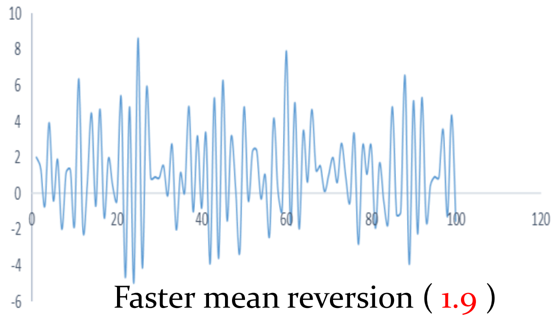


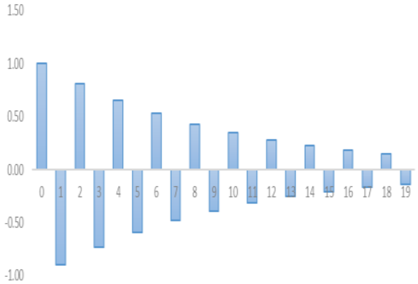


Slower mean reversion (0.1)
High Persistence

Exponentially Decaying ACF

$Y_t = 2 - 0.9Y_{t-1} + \epsilon_t$





Faster mean reversion (1.9)

Alternate +ve / -ve

The AR(2) Model

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t$$ AR (2) Model

$$\mathbb{E}[Y_t] = \phi_0 + \phi_1 \mathbb{E}[Y_{t-1}] + \phi_2 \mathbb{E}[Y_{t-2}] + \mathbb{E}[\epsilon_t]$$

$$\mu = \phi_0 + \phi_1 \mu + \phi_2 \mu + 0$$

$$\boxed{\mu = \frac{\phi_0}{(1 - \phi_1 - \phi_2)}}$$

$$cov[Y_t, Y_t] = \phi_1 cov[Y_t, Y_{t-1}] + \phi_2 cov[Y_t, Y_{t-2}] + cov[Y_t, \epsilon_t]$$

$$\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + cov[\phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \epsilon_t]$$

$$\boxed{\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2}$$ ... (1)

$$cov[Y_{t-1}, Y_t] = \phi_1 cov[Y_{t-1}, Y_{t-1}] + \phi_2 cov[Y_{t-1}, Y_{t-2}] + cov[Y_{t-1}, \epsilon_t]$$

$$\boxed{\gamma_1 = \phi_1 \gamma_0 + \phi_2 \gamma_1}$$ ... (2)

$$cov[Y_{t-2}, Y_t] = \phi_1 cov[Y_{t-2}, Y_{t-1}] + \phi_2 cov[Y_{t-2}, Y_{t-2}] + cov[Y_{t-2}, \epsilon_t]$$

$$\boxed{\gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0}$$ ... (3)

Solving 1,2,3

$$\gamma_0 = \left(\frac{1 - \phi_2}{1 + \phi_2}\right) \frac{\sigma^2}{(1 - \phi_2)^2 - {\phi_1}^2}$$

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2} \gamma_0$$

$$\gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0$$

The AR(2) Model

$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t$ AR (2) Model

$$cov[Y_t, Y_{t-l}] = \phi_1 cov[Y_{t-1}, Y_{t-l}] + \phi_2 cov[Y_{t-2}, Y_{t-l}] + \underbrace{cov[\epsilon_t, Y_{t-l}]}_{= 0 \ if \ l \neq 0}$$

$$cov[Y_t, Y_{t-l}] = \phi_1 cov[Y_{t-1}, Y_{t-l}] + \phi_2 cov[Y_{t-2}, Y_{t-l}]$$

$\boxed{\gamma_l = \phi_1 \gamma_{l-1} + \phi_2 \gamma_{l-2}}$ $l \neq 0$

Recurrence relationship

$$\frac{\gamma_l}{\gamma_0} = \phi_1 \frac{\gamma_{l-1}}{\gamma_0} + \phi_2 \frac{\gamma_{l-2}}{\gamma_0}$$

$$\boxed{\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}}$$

Yule Walker Equations

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t$$

AR (2) Model

$$\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}$$

$$\begin{bmatrix} y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} y_2 & y_1 \\ \vdots & \vdots \\ y_{n-1} & y_{n-2} \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} + \begin{bmatrix} \epsilon_3 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$(put\ l = 1) \qquad \rho_1 = \phi_1 \rho_0 + \phi_2 \rho_1$$

$$(put\ l = 2) \qquad \rho_2 = \phi_1 \rho_1 + \phi_2 \rho_0$$

$$\begin{bmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_0 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}$$

$$\boldsymbol{y} = \boldsymbol{A\phi} + \boldsymbol{\epsilon}$$

$$\boxed{\widehat{\boldsymbol{\phi}} = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\boldsymbol{y}}$$

$$\begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} 1 & \hat{\rho}_1 \\ \hat{\rho}_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} \hat{\rho}_1 \\ \hat{\rho}_2 \end{bmatrix}$$

$$\boxed{\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \frac{1}{1 - \hat{\rho}_1{}^2} \begin{bmatrix} \hat{\rho}_1 - \hat{\rho}_1 \hat{\rho}_2 \\ -\hat{\rho}_1{}^2 + \hat{\rho}_2 \end{bmatrix}}$$

Yule Walker Equations

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t$$

AR (p) Model

$$\begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p-1) \\ \rho(p) \end{pmatrix} = \begin{pmatrix} 1 & \rho(1) & \rho(2) & \dots & \rho(p-1) \\ \rho(1) & 1 & \rho(1) & \dots & \rho(p-2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho(p-2) & \rho(p-3) & \rho(p-4) & \dots & \rho(1) \\ \rho(p-1) & \rho(p-2) & \rho(p-3) & \dots & 1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_{p-1} \\ \phi_p \end{pmatrix}$$

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t$$

$$\mu = \frac{\phi_0}{1 - \phi_1 - \phi_2 - \cdots - \phi_p}$$

$$Y_t = \mu\left(1 - \phi_1 - \phi_2 - \cdots - \phi_p\right) + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t$$

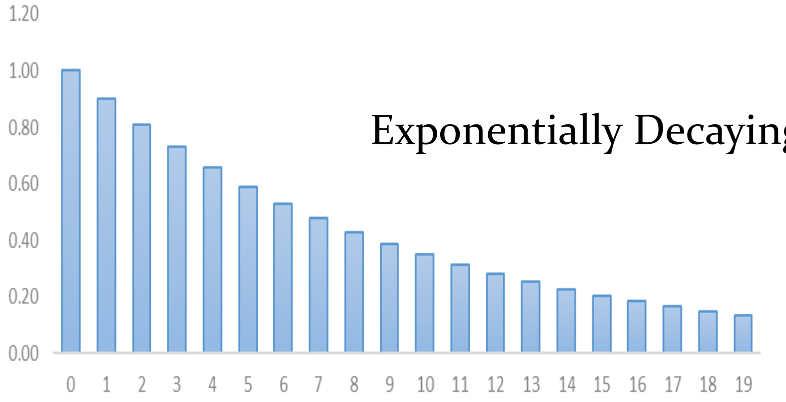
$$Y_t - \mu = \phi_1(Y_{t-1} - \mu) + \phi_2(Y_{t-2} - \mu) + \cdots + \phi_p(Y_{t-p} - \mu) + \epsilon_t$$

$$Y_t' = \phi_1(Y_{t-1}') + \phi_2(Y_{t-2}') + \cdots + \phi_p(Y_{t-p}') + \epsilon_t$$

Create $Y_t' = Y_t - \hat{\mu}$, fit AR(p) without intercept and estimate AR coefficients with Yule Walker

ACF vs PACF

$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t$ AR (1) Model



Exponentially Decaying ACF for $\phi_1 > 0$

$ACF(1) = corr(Y_t, Y_{t-1})$ ✓ Because of $Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t$

$ACF(2) = corr(Y_t, Y_{t-2})$ ✓ Because of $Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t$ and $Y_{t-1} = \phi_0 + \phi_1 Y_{t-2} + \epsilon_t$

$Y_t = \phi_0 + \phi_1 Y_{t-1} + \underbrace{\phi_2}_{0} Y_{t-2} + \epsilon_t$ AR (2) Model

ACF vs PACF

$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t$

True Model : AR (1) Model

$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t$

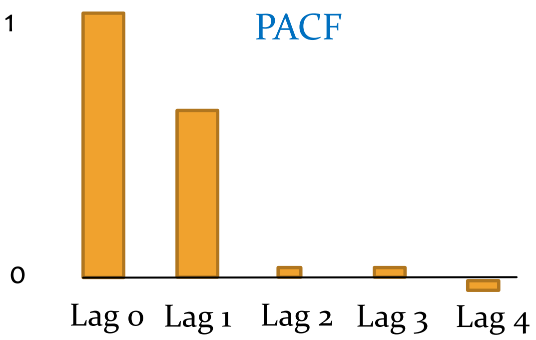
AR (2) Model

0

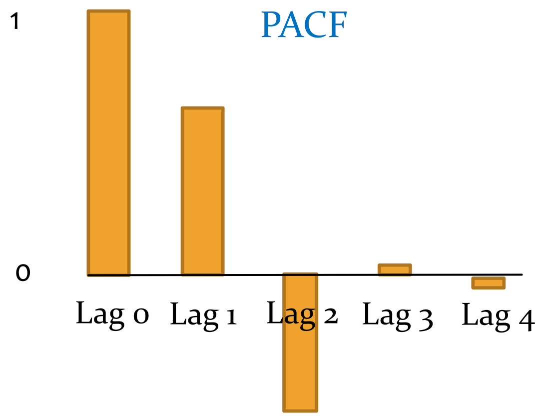
$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3} + \epsilon_t$

AR (3) Model

0



For AR (1) Process



For AR (2) Process

Determining PACF and AR order

Method 1

Fit AR (1) Model $Y_t = \phi_0 + \phi_{11} Y_{t-1} + \epsilon_t$ Estimate and Retain $\hat{\phi}_{11}$

Fit AR (2) Model $Y_t = \phi_0 + \phi_{21} Y_{t-1} + \phi_{22} Y_{t-2} + \epsilon_t$ Estimate and Retain $\hat{\phi}_{22}$

Continue until $\phi_{kk} = 0$ Conclude AR order = k-1

Method 2

$$\begin{pmatrix} \boxed{\rho(1)} \\ \rho(2) \\ \vdots \\ \rho(p-1) \\ \rho(p) \end{pmatrix} = \begin{pmatrix} \boxed{1} & \rho(1) & \rho(2) & \dots & \rho(p-1) \\ \rho(1) & 1 & \rho(1) & \dots & \rho(p-2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho(p-2) & \rho(p-3) & \rho(p-4) & \dots & \rho(1) \\ \rho(p-1) & \rho(p-2) & \rho(p-3) & \dots & 1 \end{pmatrix} \begin{pmatrix} \boxed{\phi_1} \\ \phi_2 \\ \vdots \\ \phi_{p-1} \\ \phi_p \end{pmatrix}$$

Solve and retain $\hat{\phi}_{11}$ ..it's same as $\hat{\phi}_1$

Solve and retain $\hat{\phi}_{22}$

⋮

Continue until $\phi_{kk} = 0$

Conclude AR order = k-1

The back shift (B) / lag operator (L)

$BY_t = Y_{t-1}$ $\quad$ $B^kY_t = Y_{t-k}$ $\quad$ $(1-B)Y_t = Y_t - Y_{t-1} = \Delta Y_t$

$B^2Y_t = Y_{t-2}$ $\quad$ $B^{-1}Y_t = Y_{t+1}$ $\quad$ $(1-B)BY_t = (1-B)Y_{t-1} = Y_{t-1} - Y_{t-2} = \Delta Y_{t-1}$

AR (1)

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t \longrightarrow Y_t - \phi_1 Y_{t-1} = \phi_0 + +\epsilon_t \longrightarrow (1-\phi_1 B)Y_t = \phi_0 + \epsilon_t$$

AR (2)

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t \longrightarrow \underbrace{(1-\phi_1 B - \phi_2 B^2)}_{\boldsymbol{\phi}(\boldsymbol{B})}Y_t = \phi_0 + \epsilon_t$$

Lag polynomial

AR (p)

$\boldsymbol{\phi}(\boldsymbol{B})Y_t = \phi_0 + \epsilon_t$ $\quad$ Where $\boldsymbol{\phi}(\boldsymbol{B}) = \left(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p\right)$

Stationarity of AR process

AR (1) Model

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t$$

Under Condition $|\phi_1| < 1$

$$\left.\begin{array}{l} \mu = \dfrac{\phi_0}{1-\phi_1} \\[2ex] {\sigma_Y}^2 = \dfrac{\sigma^2}{\left(1-{\phi_1}^2\right)} \\[2ex] \gamma_l = \dfrac{{\phi_1}^l}{1-{\phi_1}^2} \end{array}\right\} \text{Finite}$$

Stationarity condition $|\phi_1| < 1$

AR (p) $\boldsymbol{\phi}(\boldsymbol{B})Y_t = \phi_0 + \epsilon_t$ where $\boldsymbol{\phi}(\boldsymbol{B}) = \left(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p\right)$

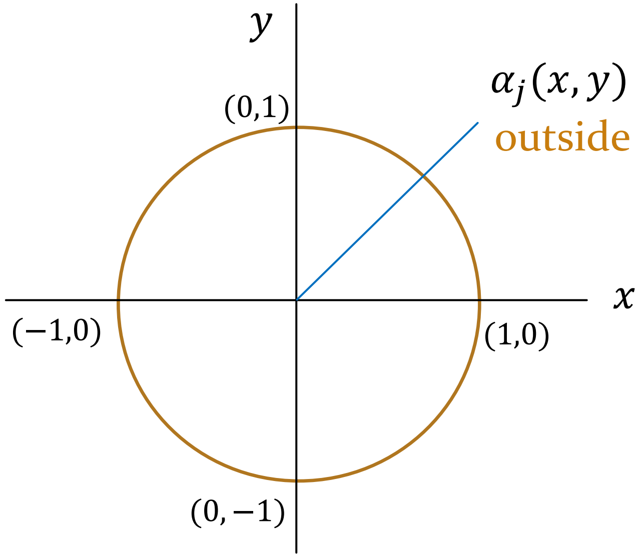
$\boldsymbol{\phi}(Z) = \left(1 - \phi_1 Z - \phi_2 Z^2 - \cdots - \phi_p Z^p\right)$ characteristic polynomial

<u>Stationarity condition</u>

For all roots (α_j) of the characteristic polynomial $\boldsymbol{\phi}(Z)$, $|\alpha_j| > 1$

$\alpha = x + iy$ $\quad i = \sqrt{-1}$

$|\alpha| > 1 \longrightarrow x^2 + y^2 > 1$



Stationarity of AR process

AR (1) $\boldsymbol{\phi}(\boldsymbol{B})Y_t = \phi_0 + \epsilon_t$ where $\boldsymbol{\phi}(\boldsymbol{B}) = (1 - \phi_1 B)$

Characteristic polynomial $\boldsymbol{\phi}(Z) = (1 - \phi_1 Z)$

$\alpha = \dfrac{1}{\phi_1}$ stationarity $|\alpha| > 1$ $\boxed{|\phi_1| < 1}$

AR (2) $\boldsymbol{\phi}(\boldsymbol{B})Y_t = \phi_0 + \epsilon_t$ where $\boldsymbol{\phi}(\boldsymbol{B}) = (1 - \phi_1 B - \phi_2 B^2)$

Characteristic polynomial $\boldsymbol{\phi}(Z) = (1 - \phi_1 Z - \phi_2 Z^2)$

$$\alpha_{1,2} = \frac{\phi_1 \pm \sqrt{{\phi_1}^2 + 4\phi_2}}{-2\phi_2}$$

$|\alpha_{1,2}| > 1$ stationarity

Infinite lag representation

AR (1) $Y_t = \epsilon_t + \phi_1 Y_{t-1}$

$$Y_t = \epsilon_t + \phi_1(\epsilon_{t-1} + \phi_1 Y_{t-2}) \quad = \epsilon_t + \phi_1\epsilon_{t-1} + {\phi_1}^2 Y_{t-2}$$

...

$$Y_t = \epsilon_t + \phi_1\epsilon_{t-1} + {\phi_1}^2\epsilon_{t-2} + \cdots$$

$$\mathbb{E}[Y_t] = 0 + \phi_1 0 + {\phi_1}^2 0 + \cdots = 0$$

$$var[Y_t] = \sigma^2 + {\phi_1}^2\sigma^2 + {\phi_1}^4\sigma^2 + \cdots$$

$$= \sigma^2(1 + {\phi_1}^2 + {\phi_1}^4 \ldots)$$

Geometric Progression converges if $|\phi_1| < 1$ $\quad var[Y_t] = \dfrac{\sigma^2}{(1 - {\phi_1}^2)}$

Linear Series $r_t = \phi_0 + \phi_1\epsilon_1 + \phi_2\epsilon_2 + \cdots$ converges if $\sum_{i=1}^{\infty}|\phi_j| < \infty$

AR (1) $(1 - \phi_1 B)Y_t = \epsilon_t$

$$Y_t = \epsilon_t(1 - \phi_1 B)^{-1}$$

$$Y_t = \epsilon_t\left(1 + \phi_1 B + {\phi_1}^2 B^2 + \cdots\right)$$

Using $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \cdots$

$$Y_t = \epsilon_t + \phi_1\epsilon_{t-1} + {\phi_1}^2\epsilon_{t-2} + \cdots$$

Infinite lag representation

AR (1) $Y_t = \epsilon_t + \phi_1 Y_{t-1}$

$Y_t = \epsilon_t + \phi_1(\epsilon_{t-1} + \phi_1 Y_{t-2}) \quad = \epsilon_t + \phi_1\epsilon_{t-1} + {\phi_1}^2 Y_{t-2}$

...

$Y_t = \epsilon_t + \phi_1\epsilon_{t-1} + {\phi_1}^2\epsilon_{t-2} + \cdots$

$\mathbb{E}[Y_t] = 0 + \phi_1 0 + {\phi_1}^2 0 + \cdots = 0$

$var[Y_t] = \sigma^2 + {\phi_1}^2\sigma^2 + {\phi_1}^4\sigma^2 + \cdots$

$= \sigma^2(1 + {\phi_1}^2 + {\phi_1}^4 \ldots)$

Geometric Progression converges if $|\phi_1| < 1$ $\quad var[Y_t] = \dfrac{\sigma^2}{(1 - {\phi_1}^2)}$

Linear Series $r_t = \phi_0 + \phi_1\epsilon_1 + \phi_2\epsilon_2 + \cdots$ converges if $\sum_{i=1}^{\infty}|\phi_j| < \infty$

AR (1) $(1 - \phi_1 B)Y_t = \epsilon_t$

$Y_t = \epsilon_t(1 - \phi_1 B)^{-1}$

$Y_t = \epsilon_t\left(1 + \phi_1 B + {\phi_1}^2 B^2 + \cdots\right)$ Using $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \cdots$

$Y_t = \epsilon_t + \phi_1\epsilon_{t-1} + {\phi_1}^2\epsilon_{t-2} + \cdots$

Goodness of fit for AR(p)

AIC (Akaike Information Criterion) $\ln(\sigma^2) + \dfrac{2p}{T}$

BIC (Bayesian Information Criterion) $\ln(\sigma^2) + \dfrac{\ln(T)p}{T}$

* σ^2 in the equations are the MLE estimate

Home work : **AIC** $= 2p - 2LL$. Show that for regression it becomes $\ln(\sigma^2) + \dfrac{2p}{T}$

Parameter Estimation for AR (p)

Yule Walker Equations

Maximum Likelihood Estimator

Ordinary Least Squares (OLS)

Forecasting

1 – step ahead forecast

$$Y_{t+1} = \phi_0 + \phi_1 Y_t + \epsilon_{t+1}$$

$$\mathbb{E}[Y_{t+1}] = \phi_0 + \phi_1 Y_t + \mathbb{E}[\epsilon_{t+1}]$$

$$\mathbb{E}[Y_{t+1}] = \phi_0 + \phi_1 Y_t$$

2 – step ahead forecast

$$Y_{t+2} = \phi_0 + \phi_1 Y_{t+1} + \epsilon_{t+2}$$

$$\mathbb{E}[Y_{t+2}] = \phi_0 + \phi_1 \mathbb{E}[Y_{t+1}] + \mathbb{E}[\epsilon_{t+2}]$$

$$\mathbb{E}[Y_{t+2}] = \phi_0 + \phi_1(\phi_0 + \phi_1 Y_t) + 0$$

$$\mathbb{E}[Y_{t+2}] = \phi_0(1 + \phi_1) + {\phi_1}^2 Y_t$$

k – step ahead forecast

$$\mathbb{E}[Y_{t+k}] = \phi_0\left(1 + \phi_1 + {\phi_1}^2 + \cdots + {\phi_1}^{k-1}\right) + {\phi_1}^k Y_t$$

$$k \to \infty \qquad \mathbb{E}[Y_{t+k}] = \frac{\phi_0}{1 - \phi_1}$$

Forecasting error variance

1 – step ahead forecast error $\quad Y_{t+1} - \mathbb{E}[Y_{t+1}] = \epsilon_{t+1}$

1 – step ahead error variance $\quad var[\epsilon_{t+1}] = \sigma^2$

2 – step ahead forecast error

$$\begin{aligned} Y_{t+2} - \mathbb{E}[Y_{t+2}] &= \phi_0 + \phi_1 Y_{t+1} + \epsilon_{t+2} - (\phi_0 + \phi_1 \mathbb{E}[Y_{t+1}]) \\ &= \epsilon_{t+2} + \phi_1 (Y_{t+1} - \mathbb{E}[Y_{t+1}]) \\ &= \epsilon_{t+2} + \phi_1 \epsilon_{t+1} \end{aligned}$$

2 – step ahead error variance $\quad var[\epsilon_{t+2} + \phi_1 \epsilon_{t+1}] = \sigma^2 (1 + {\phi_1}^2)$

k – step ahead forecast error $\quad = \epsilon_{t+k} + \phi_1 \epsilon_{t+k-1} + \cdots + {\phi_1}^{k-1} \epsilon_{t+1}$

k – step ahead error variance $\quad = \sigma^2 (1 + {\phi_1}^2 + {\phi_1}^4 + \cdots + {\phi_1}^{2(k-1)}) \quad \rightarrow \dfrac{\sigma^2}{(1 - {\phi_1}^2)} \quad if \;\; k \rightarrow \infty$

MA Models

MA(q) model

- A moving average (MA) model is one that represents Y as a linear function of previous shocks/innovations and of course the current innovation term.
- In reality, there could be processes which has a memory of previous shocks that enter explicitly in the model for future values. (e.g. stock price return today may be still affected by a major news two days ago by the firm).

MA (q) Model

$$\underbrace{Y_t = \theta_0}_{\text{Intercept}} \underbrace{- \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots - \theta_q \epsilon_{t-q}}_{\text{previous q shocks}} + \underbrace{\epsilon_t}_{\text{current shock}} \qquad \epsilon_t \sim \mathcal{N}\ (0\,, \sigma^2)$$

MA(q) model - Stationarity

- An MA model is always stationary.

MA (q) Model

$$Y_t = \theta_0 - \theta_1\epsilon_{t-1} - \theta_2\epsilon_{t-2} - \cdots - \theta_q\epsilon_{t-q} + \epsilon_t \qquad \epsilon_t \sim \mathcal{N}\ (0\,, \sigma^2)$$

$$\mathbb{E}[Y_t] = \theta_0 - \theta_1\mathbb{E}[\epsilon_{t-1}] - \theta_2\mathbb{E}[\epsilon_{t-2}] - \cdots - \theta_q\mathbb{E}[\epsilon_{t-q}] + \mathbb{E}[\epsilon_t]$$

$$\boxed{\mathbb{E}[Y_t] = \theta_0}$$

$$var[Y_t] = {\theta_1}^2 var[\epsilon_{t-1}] + {\theta_2}^2 var[\epsilon_{t-2}] + \cdots + {\theta_q}^2 var[\epsilon_{t-q}] + var[\epsilon_t]$$

$$var[Y_t] = {\theta_1}^2\sigma^2 + {\theta_2}^2\sigma^2 - \cdots + {\theta_q}^2\sigma^2 + \sigma^2$$

$$\boxed{var[Y_t] = \sigma^2\left(1 + {\theta_1}^2 + {\theta_2}^2 + \cdots + {\theta_q}^2\right)}$$

MA(q) model – Auto Covariance

MA (q) Model

$$Y_t = \theta_0 + \epsilon_t - \theta_1\epsilon_{t-1} - \theta_2\epsilon_{t-2} - \cdots - \theta_q\epsilon_{t-q} \qquad \epsilon_t \sim \mathcal{N}\ (0\,,\sigma^2)$$

$$cov[Y_t, Y_{t+h}] = cov[\theta_0 + \epsilon_t - \theta_1\epsilon_{t-1} - \theta_2\epsilon_{t-2} - \cdots - \theta_q\epsilon_{t-q}\ , \theta_0 + \epsilon_{t+h} - \theta_1\epsilon_{t+h-1} - \theta_2\epsilon_{t+h-2} - \cdots - \theta_q\epsilon_{t+h-q}]$$

$$\epsilon_t - \theta_1\epsilon_{t-1} - \theta_2\epsilon_{t-2} - \cdots - \theta_q\epsilon_{t-(q-h)} - \cdots - \theta_q\epsilon_{t-q} \qquad (Y_t)$$

$$\epsilon_{t+h} - \theta_1\epsilon_{t+h-1} - \theta_2\epsilon_{t+h-2} - \cdots - \theta_h\epsilon_{t+h-h} - \theta_{h+1}\epsilon_{t+h-(h+1)} - \cdots - \theta_{h+q}\epsilon_{t+h-q}$$ h steps ahead (Y_{t+h})

$$\underbrace{= -\theta_h\sigma^2 \qquad = \theta_1\theta_{h+1}\sigma^2 \qquad = \theta_q\theta_{h+q}\sigma^2}_{\text{q-h+1 terms}}$$

$$\boxed{\gamma_h = \sigma^2(-\theta_h + \theta_1\theta_{h+1} + \cdots + \theta_q\theta_{h+q})}$$

MA(q) model - ACF

MA (q) Model

$$Y_t = \theta_0 + \epsilon_t - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots - \theta_q \epsilon_{t-q} \qquad \epsilon_t \sim \mathcal{N}\ (0\,, \sigma^2)$$

$$\boxed{\gamma_h = \sigma^2 \left(-\theta_h + \theta_1 \theta_{h+1} + \cdots + \theta_q \theta_{h+q} \right)}$$

$$\boxed{\gamma_0 = \sigma^2 \left(1 + {\theta_1}^2 + {\theta_2}^2 + \cdots + {\theta_q}^2 \right)}$$

$$q - h + 1 \geq 1$$

$$\boxed{\rho_h = \frac{\left(-\theta_h + \theta_1 \theta_{h+1} + \cdots + \theta_q \theta_{h+q} \right)}{\left(1 + {\theta_1}^2 + {\theta_2}^2 + \cdots + {\theta_q}^2 \right)}}$$

$$h \leq q$$

(Y_t)

(Y_{t+h})

No overlap -> auto covariance 0

For MA (q) process, ACF cuts off after lag q

(e.g. for MA(1) process, ACF should drop to 0 on lag 2

Invertibility

MA (1)

$$Y_t = \epsilon_t - \theta_1 \epsilon_{t-1}$$

$$Y_t = \epsilon_t (1 - \theta_1 B)$$

$$\epsilon_t = Y_t (1 - \theta_1 B)^{-1}$$

$$\epsilon_t = Y_t (1 + \theta_1 B + {\theta_1}^2 B^2 + \cdots) \qquad |\theta_1| < 1$$

$$\epsilon_t = Y_t + \theta_1 Y_{t-1} + {\theta_1}^2 Y_{t-2} + \cdots$$

$$Y_t = \epsilon_t - \theta_1 Y_{t-1} - {\theta_1}^2 Y_{t-2} - \cdots \qquad \text{AR }(\infty)$$

MA (q)

$$Y_t = \theta_0 + \epsilon_t - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots - \theta_q \epsilon_{t-q}$$

$$Y_t = \theta_0 + \boldsymbol{\theta}(\boldsymbol{B}) \epsilon_t$$

where $\boldsymbol{\theta}(\boldsymbol{B}) = \left(1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q\right)$

is ***Invertible*** if the roots of the characteristic polynomial $\boldsymbol{\theta}(\boldsymbol{z})$ lie outside the unit circle

Stationary – Invertibility duality

Stationarity	Invertibility
• A process will be stationary if it can be represented as a linear combination of gaussian white noise terms. • If there are finite number of noise terms, no condition is needed. • If there are infinite number of terms, it must converge and we need some condition to make that happen.	• A process will be invertible if error term can be represented as the linear combination of past lags of the process. • If there are finite number of process lags, no condition is needed. • If there are infinite number of terms, it must converge and we need some condition to make that happen.
• AR process is a linear combination of infinite error terms. So to be stationary it must satisfy the condition : All roots of $\boldsymbol{\phi}(Z)$ must lie outside unit circle	• AR process is already in invertible form with finite number of process lags. (no condition is needed) *e.g.* $Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t \equiv \epsilon_t = -\phi_0 + Y_t - \phi_1 Y_{t-1}$
• MA process is a linear combination of finite error terms, so it's always stationary.	• MA process can inverted to represent error term as a combination of infinite process lags. So it needs to satisfy the condition : All roots of $\boldsymbol{\theta}(Z)$ must lie outside the unit circle

Order of MA process

MA (q) Model

$$Y_t = \theta_0 + \epsilon_t - \theta_1\epsilon_{t-1} - \theta_2\epsilon_{t-2} - \cdots - \theta_q\epsilon_{t-q}$$

ACF cuts off after lag q

PACF does not ever cut off

Estimation of MA coefficients

- MA process parameters can be estimated using maximum likelihood approach
- Depending on the order of the process, the first few error terms which are needed to kick start the process are either
  - Treated as zero (***conditional likelihood estimation***)
  - Treaded as additional parameters (***exact likelihood estimation***)

MA (2) Model

$$Y_t = \theta_0 + \epsilon_t - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2}$$

t	Y	ϵ
1	Y_1	ϵ_1
2	Y_2	ϵ_2
3	Y_3	ϵ_3
4	Y_4	ϵ_4
.	.	.
T	Y_5	ϵ_5

$$\boxed{Y_3 = \theta_0 + \epsilon_3 - \theta_1 \epsilon_2 - \theta_2 \epsilon_1}$$

$$\boxed{\epsilon_3 = -\theta_0 + Y_3 + \theta_1 \epsilon_2 + \theta_2 \epsilon_1}$$

$$\boxed{\epsilon_4 = -\theta_0 + Y_4 + \theta_1 \epsilon_3 + \theta_2 \epsilon_2}$$

$$\epsilon_t \sim \mathcal{N}\,(0\,, \sigma^2)$$

Forecasting

- Forecasting using MA model is easy because all future error terms have 0 mean and constant variance σ^2

$$Y_t = \theta_0 + \epsilon_t - \theta_1\epsilon_{t-1} - \theta_2\epsilon_{t-2}$$

MA (2) Model

1 step ahead

$$\mathbb{E}[Y_{t+1}] = \theta_0 + \underset{0}{\underbrace{\mathbb{E}[\epsilon_{t+1}]}} - \underbrace{\theta_1\mathbb{E}[\epsilon_t] - \theta_2\mathbb{E}[\epsilon_{t-1}]}_{known\ at\ t}$$

$$\boxed{\mathbb{E}[Y_{t+1}] = \theta_0 - \theta_1\epsilon_t - \theta_2\epsilon_{t-1}}$$

$$\boxed{var[Y_{t+1}] = var[\epsilon_{t+1}] = \sigma^2}$$

2 step ahead

$$\mathbb{E}[Y_{t+2}] = \theta_0 + \underset{0}{\underbrace{\mathbb{E}[\epsilon_{t+2}]}} - \theta_1\underset{0}{\underbrace{\mathbb{E}[\epsilon_{t+1}]}} - \theta_2\underbrace{\mathbb{E}[\epsilon_t]}_{known\ at\ t}$$

$$\boxed{\mathbb{E}[Y_{t+2}] = \theta_0 - \theta_2\epsilon_t}$$

$$\boxed{var[Y_{t+2}] = var[\epsilon_{t+2}] + {\theta_1}^2 var[\epsilon_{t+1}] = \sigma^2(1 + {\theta_1}^2)}$$

3 step ahead and beyond

$$\boxed{\mathbb{E}[Y_{t+3}] = \theta_0}$$

$$\boxed{var[Y_{t+3}] = \sigma^2(1 + {\theta_1}^2 + {\theta_2}^2)}$$

ARMA Models

ARMA(p , q) model

- To obtain more flexibility in the model, we want to have both AR and MA terms in the model. This is called as ARMA (p, q) models where p is the order of AR and q is the order of MA.
- The other reason why we want to have ARMA models is parsimony in parameters.

ARMA (p , q) Model

$$\underbrace{Y_t = \phi_0}_{\text{Intercept}} + \underbrace{\phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p}}_{\text{p} - \text{lags (AR)}} - \underbrace{\theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots - \theta_q \epsilon_{t-q}}_{\text{q} - \text{lags (MA)}} + \underbrace{\epsilon_t}_{\text{shock}} \qquad \epsilon_t \sim \mathcal{N}\ (0\,, \sigma^2)$$

ARMA(p , q) model - representation

- There are 3 ways in which we can represent an ARMA model.

1

$$Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} + \cdots - \phi_p Y_{t-p} = \phi_0 - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots - \theta_q \epsilon_{t-q} + \epsilon_t$$

$$Y_t \boldsymbol{\phi}(\boldsymbol{B}) = \phi_0 + \epsilon_t \boldsymbol{\theta}(\boldsymbol{B})$$

Where $\boldsymbol{\phi}(\boldsymbol{B}) = \left(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p\right)$ and $\boldsymbol{\theta}(\boldsymbol{B}) = \left(1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q\right)$

$$\frac{\boldsymbol{\theta}(\boldsymbol{B})}{\boldsymbol{\phi}(\boldsymbol{B})} = 1 + \psi_1 B + \psi_2 B^2 + \cdots$$

$$\frac{\boldsymbol{\phi}(\boldsymbol{B})}{\boldsymbol{\theta}(\boldsymbol{B})} = 1 - \pi_1 B - \pi_2 B^2 + \cdots$$

$$Y_t \boldsymbol{\phi}(\boldsymbol{B}) = \phi_0 + \epsilon_t \boldsymbol{\theta}(\boldsymbol{B})$$

Divide by $\boldsymbol{\theta}(\boldsymbol{B})$

Divide by $\boldsymbol{\phi}(\boldsymbol{B})$

2

$$Y_t = \frac{\phi_0}{1 - \theta_1 - \theta_2 - \cdots - \theta_q} + \pi_1 Y_{t-1} + \pi_2 Y_{t-2} + \cdots + \epsilon_t$$

AR representation

3

$$Y_t = \frac{\phi_0}{1 - \phi_1 - \phi_2 - \cdots - \phi_p} + \epsilon_t + \psi_1 \epsilon_{t-1} + \psi_2 \epsilon_{t-2} + \cdots$$

MA representation

ARMA(p , q) model – stationarity & invertibility

- ARMA model is stationary if the AR part is ***stationary*** (roots of $\boldsymbol{\phi}(\boldsymbol{Z})$ must lie outside unit circle)
- ARMA model is invertible if the MA part is ***invertible*** (roots of $\boldsymbol{\theta}(\boldsymbol{Z})$ must lie outside unit circle)

$$Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} + \cdots - \phi_p Y_{t-p} = \phi_0 - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots - \theta_q \epsilon_{t-q} + \epsilon_t$$

$Y_t \boldsymbol{\phi}(\boldsymbol{B}) = \phi_0 + \epsilon_t \boldsymbol{\theta}(\boldsymbol{B})$ Where $\boldsymbol{\phi}(\boldsymbol{B}) = \left(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p\right)$ and $\boldsymbol{\theta}(\boldsymbol{B}) = \left(1 - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q\right)$

Stationarity Condition : Roots of $\boldsymbol{\phi}(\boldsymbol{Z})$ must lie outside of unit circle

Invertibility Condition : Roots of $\boldsymbol{\theta}(\boldsymbol{Z})$ must lie outside of unit circle

ARMA(1 , 1) model – Mean, Variance, Auto covariance

- The Mean, Variance and Covariance for ARMA(1,1) can be calculated as follows.

ARMA (1 , 1)

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t - \theta_1 \epsilon_{t-1}$$

$$\mathbb{E}[Y_t] = \phi_0 + \phi_1 \mathbb{E}[Y_{t-1}]$$

$$\boxed{\mathbb{E}[Y_t] = \frac{\phi_0}{1 - \phi_1}}$$

$$var[Y_t] = {\phi_1}^2 var[Y_{t-1}] + var[\epsilon_t] + {\theta_1}^2 var[\epsilon_{t-1}] - 2\phi_1\theta_1 cov(Y_{t-1}, \epsilon_{t-1})$$

$${\sigma_Y}^2 = {\phi_1}^2 {\sigma_Y}^2 + \sigma^2 + {\theta_1}^2 \sigma^2 - 2\phi_1\theta_1\sigma^2$$

$$\boxed{\gamma_0 = {\sigma_Y}^2 = \frac{\sigma^2\left(1 + {\theta_1}^2 - 2\phi_1\theta_1\right)}{1 - {\phi_1}^2}}$$

$$cov[Y_t, Y_{t-1}] = \phi_1 cov[Y_{t-1}, Y_{t-1}] + cov[\epsilon_t, Y_{t-1}] - \theta_1 cov[\epsilon_{t-1}, Y_{t-1}]$$

$$\gamma_1 = \phi_1\gamma_0 + 0 - \theta_1\sigma^2$$

$$\boxed{\rho_1 = \frac{\gamma_1}{\gamma_0} = \phi_1 - \frac{\theta_1\sigma^2}{\gamma_0}}$$

$$\boxed{\rho_h = \phi_1 \rho_{h-1}}$$ for h >= 2

ARMA(p , q) model – Estimation

- Parameters of ARMA can be estimated using maximum likelihood approach (conditional or exact)

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \epsilon_t - \theta_1 \epsilon_{t-1}$$

ARMA (1 , 1)

$$\epsilon_t = Y_t - \phi_0 - \phi_1 Y_{t-1} + \theta_1 \epsilon_{t-1} \quad \sim \mathcal{N}\,(0\,, \sigma^2)$$

ARMA(p , q) – Order Determination

- There are several methods in literature to determine the order of ARMA model i.e. find p and q.
- Some popular methods are
  - AIC / BIC
  - EACF (Extended Auto Correlation Function)
  - SCAN

Non Stationary Models

Stationary Models in Finance

- Interest rates are modeled using stationary processes such as AR(1)

$$r_t = \mu + \alpha r_{t-1} + \epsilon_t \qquad \epsilon_t \sim \mathcal{N}\,(0, {\sigma_t}^2) \qquad \sigma_t = \kappa {r_{t-1}}^{\gamma}$$

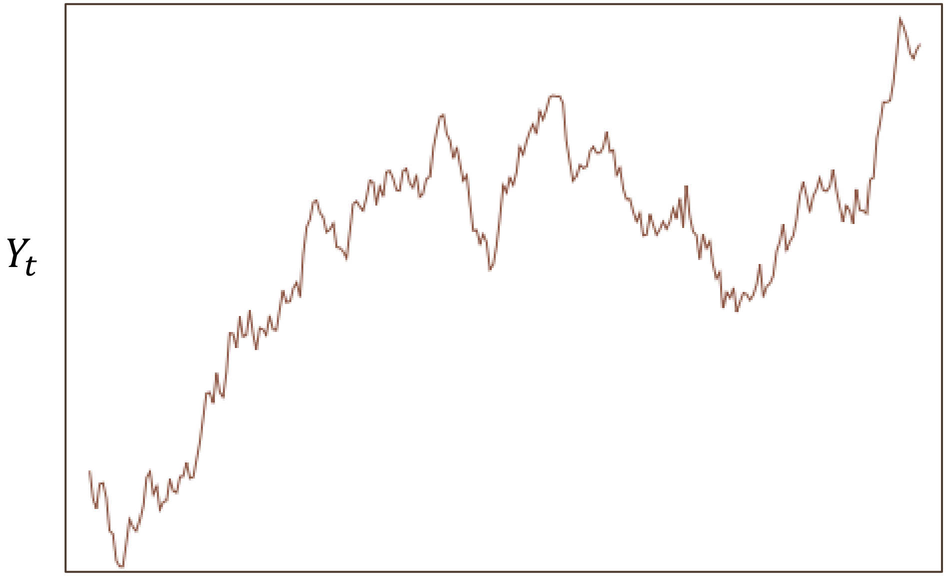
$\gamma = 0\,, \qquad \textit{Vasicek Model}$

$\gamma = {}^{1}/_{2}\,, \qquad \textit{CIR Model}$

Random Walk Model

- Random Walk model is extremely popular in Finance which are used to model non-stationary quantities such as equity prices, commodity prices and FX rates.
- The mean, variance and covariance are functions of time.

$Y_t = \mu + Y_{t-1} + \epsilon_t$ Random Walk $\epsilon_t \sim \mathcal{N}\ (0, \sigma^2)$



$Y_t - Y_{t-1} = \mu + \epsilon_t$ Increment

$\Delta Y_t = \mu + \epsilon_t$

$\Delta Y_t \sim \mathcal{N}\ (\mu, \sigma^2)$

Random Walk Model

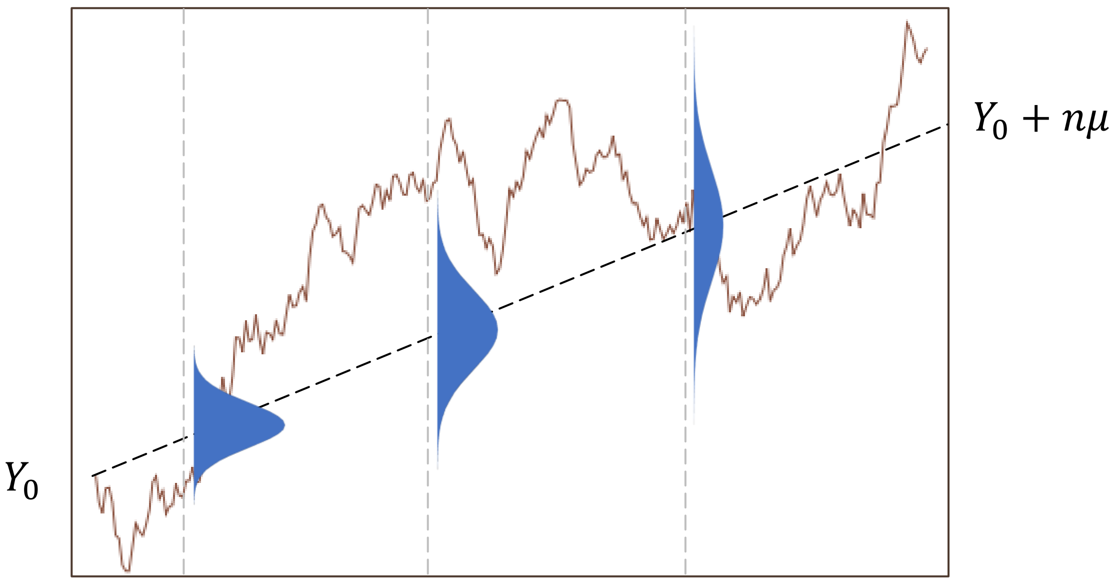
- The mean, variance and covariance vary linearly with time.

$$\underbrace{Y_n = Y_0 + \Delta Y_1 + \Delta Y_2 + \cdots + \Delta Y_n}$$

$$\Delta Y_i \sim \mathcal{N}\ (\mu, \sigma^2)$$

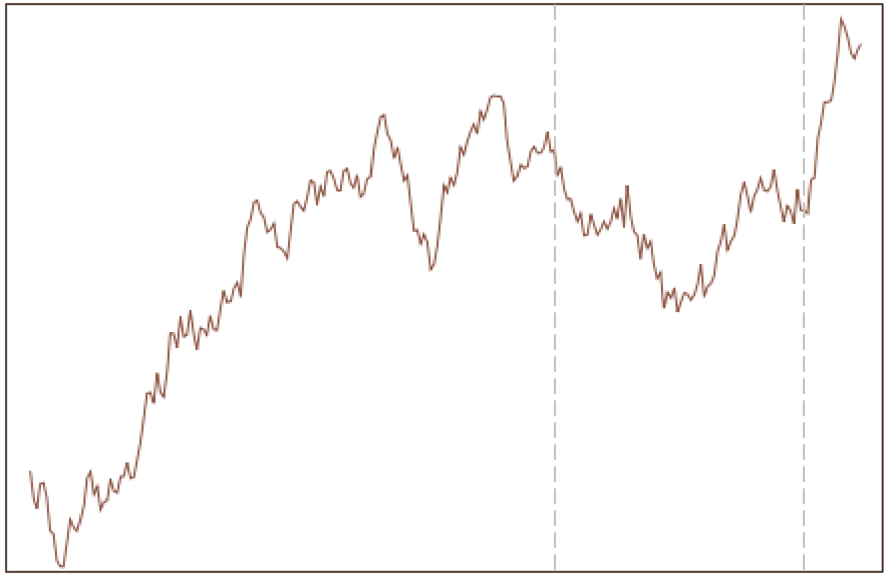
$$\sum \Delta Y_i \sim \mathcal{N}\ (n\mu, n\sigma^2)$$

$$Y_n = \mathcal{N}\ (Y_0 + n\mu, n\sigma^2)$$



Random Walk Model

- The mean, variance and covariance vary linearly with time.



$$Y_n = Y_m + \Delta Y$$

$$\Delta Y \sim \mathcal{N}\left((n-m)\mu, (n-m)\sigma^2\right)$$

$$cov[Y_n, Y_m] = cov[Y_m + \Delta Y, Y_m] = cov\,[Y_m, Y_m] + cov\,[\Delta Y, Y_m]$$

Increment independent of past
$cov\,[\Delta Y, Y_m] = 0$

$$cov[Y_n, Y_m] = var\,[Y_m] = m\sigma^2$$

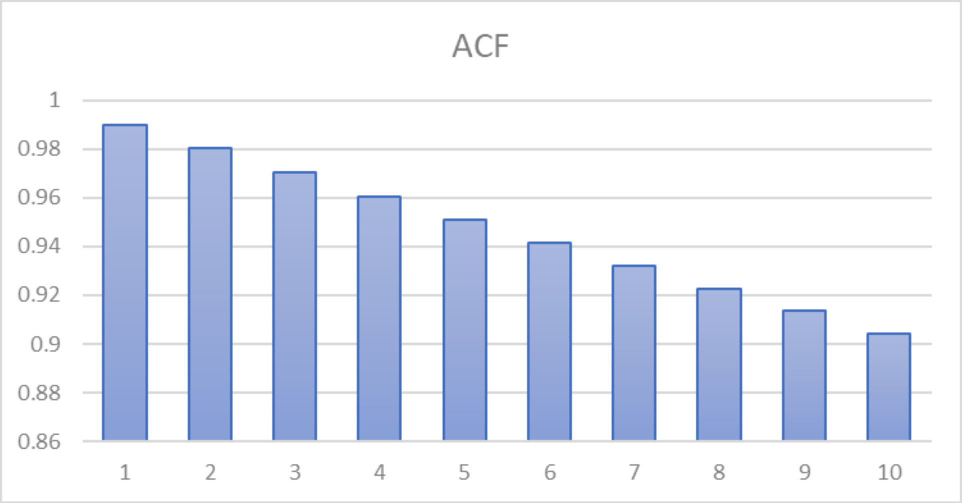
$$\rho = \frac{cov[Y_n, Y_m]}{\sqrt{var\,[Y_m] var\,[Y_n]}} = \frac{m\sigma^2}{\sqrt{m\sigma^2 n\sigma^2}} = \sqrt{\frac{m}{n}}$$

Random Walk Model - ACF

- ACF of a non-stationary process is close to 1 for several lags.

$$\rho = \sqrt{\frac{m}{n}}$$

$$\rho_1 = \sqrt{\frac{n-1}{n}} \qquad \rho_2 = \sqrt{\frac{n-2}{n}} \qquad \cdots \qquad \rho_k = \sqrt{\frac{n-k}{n}} \qquad \rho_k \approx 1 \text{ if } n \gg k$$



close to 1 &
slow decay

Unit root Non stationary processes

- The random walk processes we looked at is an example of a unit root non stationary process.
- As the name suggests, the coefficient on the lag term is 1 (hence unit root)
- Some more properties of Random Walk Model
    - Forecast of Y is equal to most recent value (Martingale)
    - Mean and Variance increase linearly with time
    - Unconditional Mean and Variance are infinity
    - The process has infinite memory (remembers all past shocks)

$$Y_t = \mu + Y_{t-1} + \epsilon_t$$

$$Y_{t+\tau} = Y_t + \Delta Y \qquad \Delta Y \sim \mathcal{N}\,(\mu\tau, \sigma^2\tau)$$

$$\mathbb{E}[Y_{t+\tau}] = Y_t + \mu\tau$$

Martingale if $\mu = 0$

$$Y_t = \mu + Y_{t-1} + \epsilon_t$$

$$Y_t = 2\mu + Y_{t-2} + \epsilon_t + \epsilon_{t-1}$$

$$Y_t = t\mu + Y_0 + \epsilon_t + \epsilon_{t-1} + \epsilon_{t-2} + \cdots + \epsilon_1$$

Infinite memory of past shocks

Trend stationary vs Unit root non stationary

- Both trend stationary and unit root stationary show explosive behavior
- But the difference is a trend stationary time series has a trending mean but constant variance and we can detrend to get a stationary time series
- Whereas in case of a unit root non stationary time series such as a random walk process both mean and variance increase with time. To make it stationary we use 'differencing'

Trend Stationary

$$Y_t = \mu + \beta t + \epsilon_t$$

$$Y_t \sim \mathcal{N}(\mu + \beta t, \sigma^2)$$

Regress Y_t with t , obtain β , then **detrend** $Y_t' = Y_t - \beta t$

Y_t' is stationary

Random Walk

$$Y_t = \mu + Y_{t-1} + \epsilon_t$$

$$Y_t \sim \mathcal{N}(\mu t, \sigma^2 t)$$

(differencing) $\Delta Y_t = Y_t - Y_{t-1}$

$\Delta Y_t = \mu + \epsilon_t$ is stationary

$$\Delta Y_t = (1 - B) Y_t$$

ARIMA (p,d,q)

- An ARIMA (p,d,q) stands for Autoregressive Integrated Moving average process, where d represents the number of unit roots of the AR polynomial

$Y_t \sim$ ARIMA $(p, 1, q)$

$(1 - B)Y_t \sim$ ARMA (p, q)

$Y_t = \mu + Y_{t-1} + \epsilon_t$ I (1) process

ϵ_t I (0) process

$Y_t \sim$ ARIMA $(p, 2, q)$

$(1 - B)^2 Y_t \sim$ ARMA (p, q)

Dicky Fuller Test for Unit root non-stationary

- Unit root non-stationary can be checked by Dicky Fuller test

$$Y_t = \rho Y_{t-1} + \epsilon_t$$

$$Y_t - Y_{t-1} = \rho Y_{t-1} - Y_{t-1} + \epsilon_t$$

$$\Delta Y_t = (\rho - 1) Y_{t-1} + \epsilon_t$$

$$\Delta Y_t = \delta Y_{t-1} + \epsilon_t$$

$$t - stat = \frac{\hat{\delta}}{s.e.(\hat{\delta})}$$

$H_0 : \delta = 0$ Process is unit root non-stationary

$H_a : \delta < 0$ Process is stationary

Augmented Dicky Fuller test

- T-stat follows Dicky Fuller distribution and critical values depend on model version and sample size.

$$\Delta Y_t = \delta Y_{t-1} + \epsilon_t$$ 1... (no intercept)

$$\Delta Y_t = \alpha_0 + \delta Y_{t-1} + \epsilon_t$$ 2... (with intercept)

$$\Delta Y_t = \alpha_0 + \alpha_1 t + \delta Y_{t-1} + \epsilon_t$$ 3... (with intercept and trend)

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \delta_1 \Delta Y_{t-1} + \delta_2 \Delta Y_{t-2} + \cdots + \delta_p \Delta Y_{t-p} + \epsilon_t$$

Augmented Dicky Fuller Test

$H_0 : \gamma = 0$ Unit root non-stationary

$H_a : \gamma < 0$

Critical values for Dickey–Fuller *t*-distribution.

	Without trend		With trend	
Sample size	1%	5%	1%	5%
T = 25	−3.75	−3.00	−4.38	−3.60
T = 50	−3.58	−2.93	−4.15	−3.50
T = 100	−3.51	−2.89	−4.04	−3.45
T = 250	−3.46	−2.88	−3.99	−3.43
T = 500	−3.44	−2.87	−3.98	−3.42
T = ∞	−3.43	−2.86	−3.96	−3.41

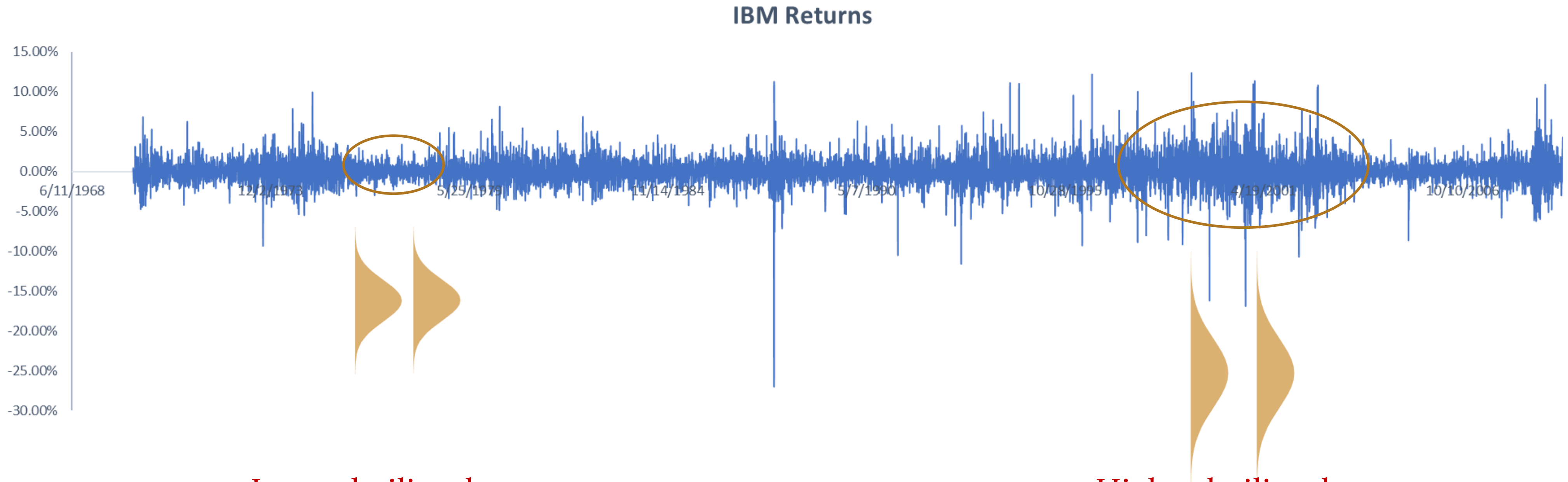
Conditional Volatility Models

Characteristics of volatility

- Volatility Clustering - High volatilities and Low volatilities form separate clusters
- Continuous evolution - Volatilities evolve in a continuous fashion. Jumps are rare.
- Finite range - Volatility is not infinity, it varies within a range. Loosely speaking, it's stationary.

$$r_t = \phi_0 + \phi_1 r_{t-1} + \epsilon_t \qquad \epsilon_t \sim \mathcal{N}(0, \sigma^2) \qquad \epsilon_t \sim \mathcal{N}(0, {\sigma_t}^2)$$

Conditional volatility



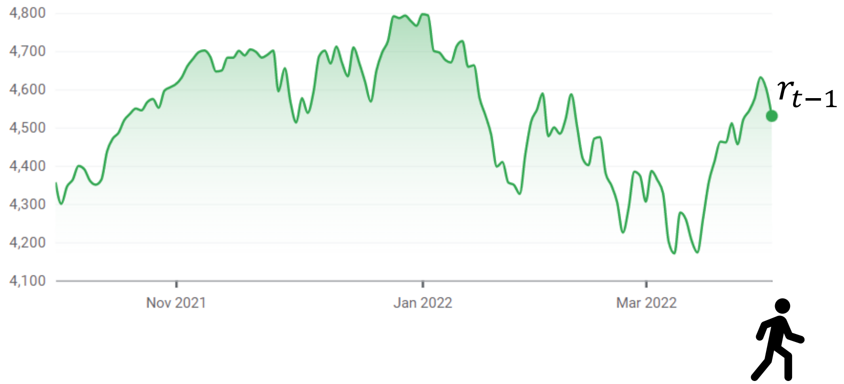
Low volatility cluster

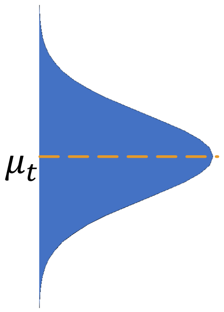
High volatility cluster

Volatility Models

- Two types of volatility models exist
  - Deterministic Volatility Models
  - Stochastic Volatility Models

$$r_t | r_{t-1} \sim \mathcal{N}(\phi_0 + \phi_1 r_{t-1}, {\sigma_t}^2)$$





$$r_t = \phi_0 + \phi_1 r_{t-1} + \epsilon_t$$

$$\mu_t = \phi_0 + \phi_1 r_{t-1}$$

$$\epsilon_t \sim \mathcal{N}(0, {\sigma_t}^2)$$

Deterministic

$${\sigma_t}^2 = \alpha_0 + \alpha_1 {\epsilon_{t-1}}^2 + \beta_1 {\sigma_{t-1}}^2$$

GARCH (1,1)

Stochastic

$${\sigma_t}^2 = \alpha_0 + \beta_1 {\sigma_{t-1}}^2 + \xi \sqrt{\sigma_t} Z_t$$

$$Z \sim \mathcal{N}(0,1)$$

e.g. Heston , SABR

ARCH Model

- ARCH(m) model : ${\sigma_t}^2 = \alpha_0 + \alpha_1 {\epsilon_{t-1}}^2 + \cdots + \alpha_m {\epsilon_{t-m}}^2$

$${\sigma_t}^2 = \alpha_0 + \alpha_1 {\epsilon_{t-1}}^2 + \cdots + \alpha_m {\epsilon_{t-m}}^2$$

Proxy for ${\sigma_{t-1}}^2$ Proxy for ${\sigma_{t-m}}^2$

$$\epsilon_t \sim \mathcal{N}(0, {\sigma_t}^2)$$

$$\mathbb{E}[\epsilon_t] = {\sigma_t}^2$$

ARCH(1) : ${\sigma_t}^2 = \alpha_0 + \alpha_1 {\epsilon_{t-1}}^2$

$${\sigma_{unconditional}}^2 = \alpha_0 + \alpha_1 {\sigma_{unconditional}}^2$$

$${\sigma_{unconditional}}^2 = \alpha_0 / (1 - \alpha_1)$$

ARCH(m) : ${\sigma_t}^2 = \alpha_0 + \alpha_1 {\epsilon_{t-1}}^2 + \cdots + \alpha_m {\epsilon_{t-m}}^2$

$${\sigma_{unconditional}}^2 = \alpha_0 / (1 - \alpha_1 - \alpha_2 - \cdots - \alpha_m)$$

we need $\alpha_1 + \alpha_2 + \cdots + \alpha_m < 1$

GARCH Model

- GARCH(p , q) model : $\sigma_t{}^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}{}^2 + \cdots + \alpha_p \epsilon_{t-p}{}^2 + \beta_1 \sigma_{t-1}{}^2 + \cdots + \beta_q \sigma_{t-q}{}^2$

$$\sigma_t{}^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}{}^2 + \cdots + \alpha_p \epsilon_{t-p}{}^2 + \beta_1 \sigma_{t-1}{}^2 + \cdots + \beta_q \sigma_{t-q}{}^2$$

GARCH(1,1) : $\sigma_t{}^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}{}^2 + \beta_1 \sigma_{t-1}{}^2$

$$\mathbb{E}[\sigma_t{}^2] = \alpha_0 + \alpha_1 \mathbb{E}[\epsilon_{t-1}{}^2] + \beta_1 \mathbb{E}[\sigma_{t-1}{}^2]$$

$$\mathbb{E}[\sigma_t{}^2] = \alpha_0 + \alpha_1 \mathbb{E}[\epsilon_{t-1}{}^2] + \beta_1 \mathbb{E}[\sigma_{t-1}{}^2]$$

$$\sigma_{unconditional}{}^2 = \frac{\alpha_0}{(1 - \alpha_1 - \beta_1)} = \omega_0$$

$$\alpha_1 + \beta_1 < 1$$

GARCH(1,1) : $\sigma_t{}^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}{}^2 + \beta_1 \sigma_{t-1}{}^2$

$$\sigma_t{}^2 = \omega_0 (1 - \alpha_1 - \beta_1) + \alpha_1 \epsilon_{t-1}{}^2 + \beta_1 \sigma_{t-1}{}^2$$

$$\alpha_1 + \beta_1$$

Speed of mean reversion

persistence

GARCH Model - Forecasting

- GARCH(p , q) model : $\sigma_t{}^2 = \alpha_0 + \alpha_1\epsilon_{t-1}{}^2 + \cdots + \alpha_p\epsilon_{t-p}{}^2 + \beta_1\sigma_{t-1}{}^2 + \cdots + \beta_q\sigma_{t-q}{}^2$

GARCH(1,1) : $\sigma_t{}^2 = \alpha_0 + \alpha_1\epsilon_{t-1}{}^2 + \beta_1\sigma_{t-1}{}^2$

$\sigma_0{}^2$ $\quad\sigma_{k-2}{}^2$, $\epsilon_{k-2}{}^2$ $\quad\sigma_{k-1}{}^2$, $\epsilon_{k-1}{}^2$ $\quad\sigma_k{}^2$

$0 \qquad k-2 \qquad k-1 \qquad k$

$$\sigma_k{}^2|\sigma_0{}^2 = \alpha_0 + (\alpha_1+\beta_1)\sigma_{k-1}{}^2 \quad = \alpha_0 + \eta\sigma_{k-1}{}^2$$

$$= \alpha_0 + \eta(\alpha_0 + \eta\sigma_{k-2}{}^2) \quad = \alpha_0(1+\eta) + \eta^2\sigma_{k-2}{}^2$$

$$\cdots$$

$$= \alpha_0(1+\eta+\cdots+\eta^k) + \eta^k\sigma_0{}^2 \quad = \alpha_0\left(\frac{1-\eta^k}{1-\eta}\right) + \eta^k\sigma_0{}^2$$

$$if\ \ k \to \infty, \sigma_k{}^2|\sigma_0{}^2 \to \frac{\alpha_0}{1-\eta} = \omega_0$$ (Unconditional Variance)

IGARCH / EWMA Model

- Special case of GARCH(1,1) with persistence = 1 called the IGARCH(1,1) model

GARCH(1,1) : $\sigma_t{}^2 = \alpha_0 + \alpha_1\epsilon_{t-1}{}^2 + \beta_1\sigma_{t-1}{}^2$ $\quad$ $if\ \ \alpha_1 + \beta_1 = 1\,,$ $\quad$ IGARCH(1,1)

IGARCH(1,1) : $\sigma_t{}^2 = \alpha_0 + (1-\lambda)\epsilon_{t-1}{}^2 + \lambda\sigma_{t-1}{}^2$

EWMA : $\sigma_t{}^2 = (1-\lambda)\epsilon_{t-1}{}^2 + \lambda\sigma_{t-1}{}^2$

$$= (1-\lambda)\epsilon_{t-1}{}^2 + \lambda((1-\lambda)\epsilon_{t-2}{}^2 + \lambda\sigma_{t-2}{}^2) \qquad = (1-\lambda)\epsilon_{t-1}{}^2 + \lambda(1-\lambda)\epsilon_{t-2}{}^2 + \lambda^2\sigma_{t-2}{}^2$$

…

$$= (1-\lambda)\{\lambda^0\epsilon_{t-1}{}^2 + \lambda(1-\lambda)\epsilon_{t-2}{}^2 + \cdots\} \qquad = (1-\lambda)\sum \lambda^{i-1}\epsilon_{t-i}{}^2$$

GARCH estimation

- Estimation of GARCH parameters can be done using the ususal MLE approach. But there are two concerns
  - Too much data might include outliers in the past that will influence forecast today
  - Too less data will bring uncertainty in parameter estimates

$$r_t = \phi_0 + \phi_1 r_{t-1} + \epsilon_t \qquad \epsilon_t \sim \mathcal{N}(0, {\sigma_t}^2)$$

$$\mu_t = \phi_0 + \phi_1 r_{t-1} \qquad {\sigma_t}^2 = \alpha_0 + \alpha_1 {\epsilon_{t-1}}^2 + \beta_1 {\sigma_{t-1}}^2$$

(MLE steps)

1. $\epsilon_t = r_t - \mu_t$

2. $L(\epsilon_t) = \dfrac{1}{\sigma_t\sqrt{2\pi}} \exp\left\{-\dfrac{1}{2}\left(\dfrac{\epsilon_t}{\sigma_t}\right)^2\right\}$ $\qquad or \quad LL(\epsilon_t) \cong -\dfrac{1}{2}\ln\left({\sigma_t}^2 + \left(\dfrac{\epsilon_t}{\sigma_t}\right)^2\right)$

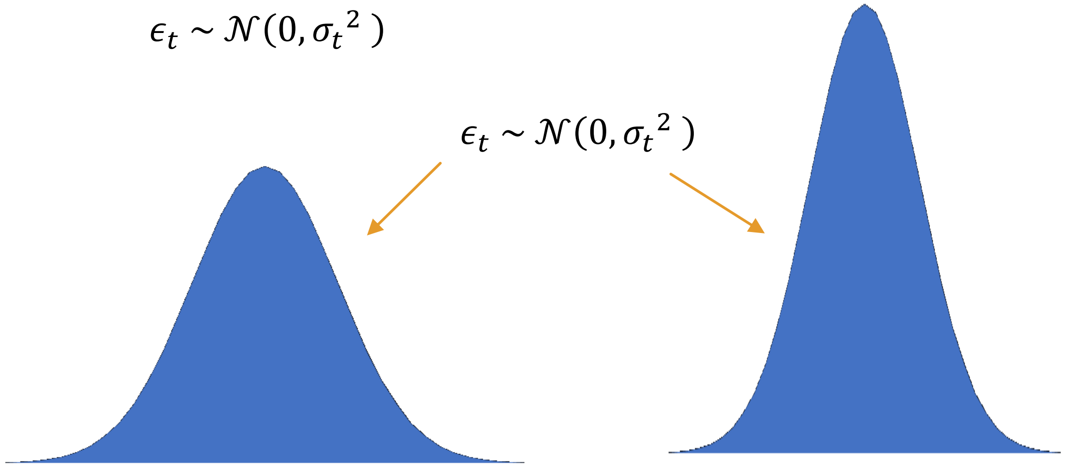
Return Distributions

Marginal Distribution & Conditional Distribution

- The distribution of noise term is called Unconditional or *Marginal Distribution*
- The distribution of standardized noise term is called *Conditional distribution*
- Unconditional distributions are heavy tailed compared to Gaussian

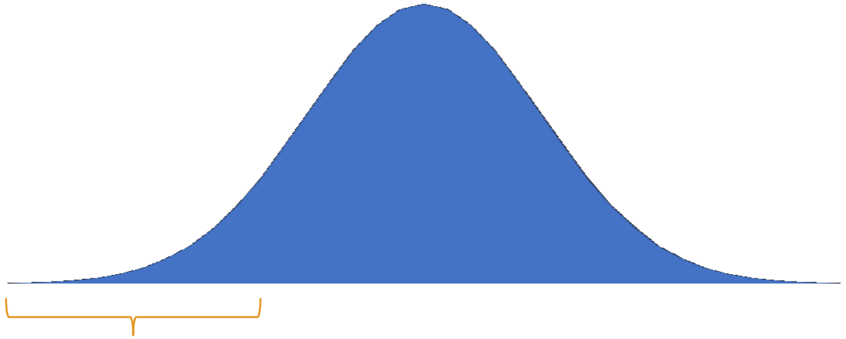
$r_t = \mu_t + \epsilon_t$

$\epsilon_t \sim \mathcal{N}(0, {\sigma_t}^2)$



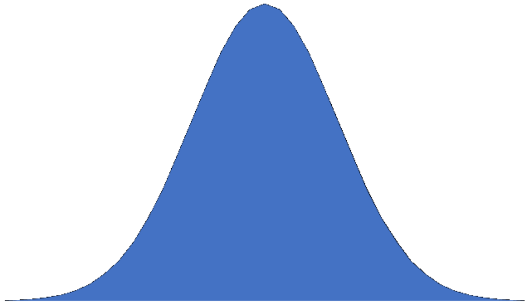
$at\ t_1$ $at\ t_2$

$Marginal\ Distribution\ (not\ gaussian)$



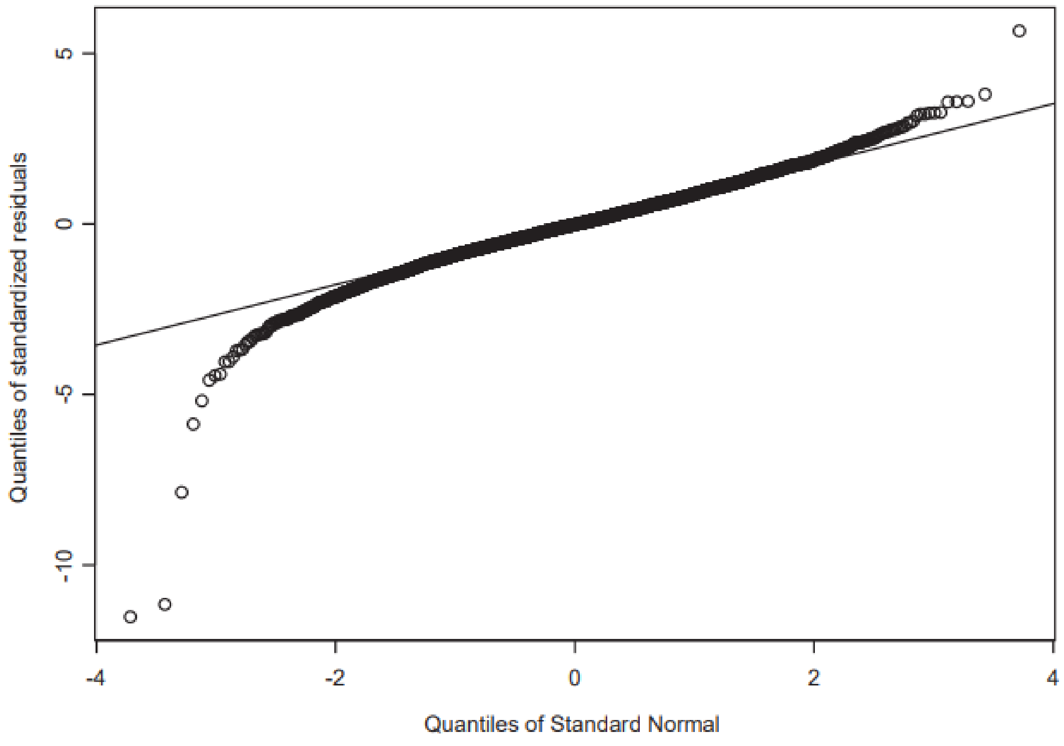
$heavy\ tails$

$Z = \epsilon_t / \sigma_t \sim \mathcal{N}(0,1)$



Testing for Normality – QQ Plot

- Normality can be visually tested using QQ plot
- Statistical tests such as Shapiro-Wilk’s test, Anderson Darling test etc can also be used but the tests lack reliability



Student T- Distribution

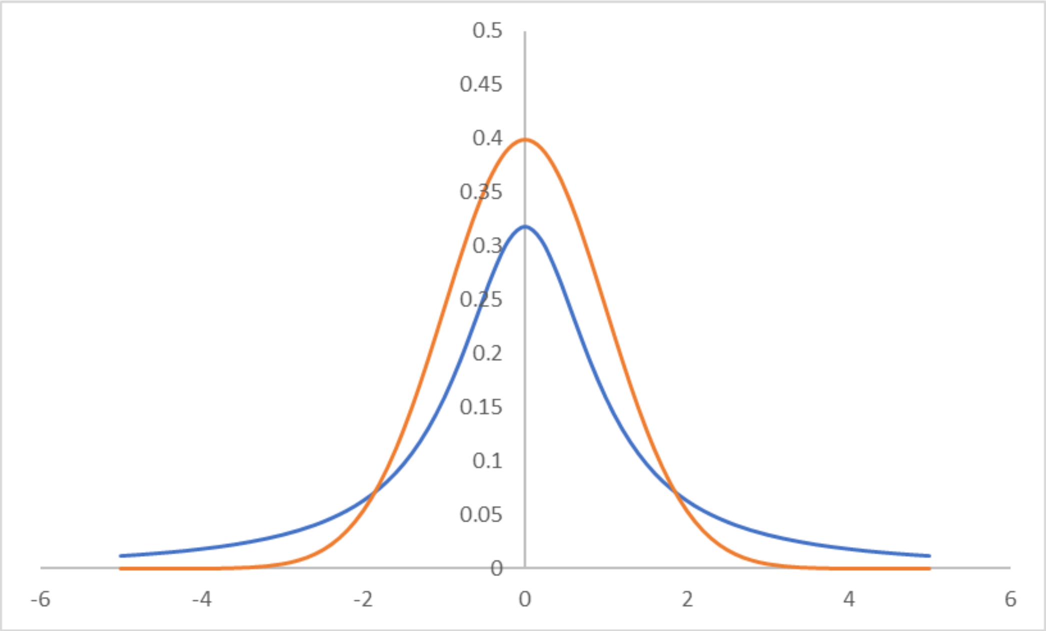
- T- distribution allows for fatter tails but is symmetric

$$f_x(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\,\sigma\,\sqrt{\pi\,(\nu-2)}}\left[1+\frac{(x-\mu)^2}{(\nu-2)\,\sigma^2}\right]^{-\left(\frac{\nu+1}{2}\right)}$$

μ – location parameter

σ – dispersion parameter

ν – degree of freedom



Normal Inverse Gaussian Distribution (NIG)

- NIG distribution allows for skewness and tailedness

$$f_x(x) = \frac{\delta\,\alpha}{\pi\, q(x)} \exp\left(p(x)\right)\, K_1\left(\alpha\, q(x)\right)$$

K_1 – modified Bessel function of second kind of order 1

$$p(x) = \delta\sqrt{\alpha^2 - \beta^2} + \beta\,(x - \mu)$$

$$q(x) = \sqrt{\delta^2 + (x - \mu)^2}$$

α , β control the shape of the density

$$\text{skewness} = \frac{\beta}{\alpha}\left(1 + \delta\sqrt{\alpha^2 - \beta^2}\right)^{-1/2}$$

$$\text{fat tailedness} = \left(1 + \delta\sqrt{\alpha^2 - \beta^2}\right)^{-1/2}$$

Cointegration

Co-integration

- If both Y_t and X_t are I(1), but a linear combination of them is I(0), then we say they are co-integrated.
- In general if there are K time series variables $(X_1, X_2, \dots X_k)$ and they are all I(d) and if a linear combination of them is I(f) such that f < d, then we say they are co-integrated.

$Y_t = \phi_0 + Y_{t-1} + \epsilon_t$ I(1) – one unit root

$\Delta Y_t = \phi_0 + \epsilon_t$ I(0) – stationary

$(1 - B)^2 Y_t = \phi_0 + \epsilon_t$ I(2) – two unit roots

$\Delta(\Delta Y_t) = \phi_0 + \epsilon_t$ I(0) – stationary

$X_t \; is$ I(1) $and \; Y_t \; is$ I(1)

$$Y_t = \alpha + \beta X_t + \epsilon$$

$$\underbrace{Y_t - \alpha - \beta X_t}_{\mathrm{I}(1)} = \underbrace{\epsilon}_{\mathrm{I}(0)}$$

not possible

$$\Delta Y_t = \alpha' + \beta' \Delta X_t + \epsilon$$

possible

Co-integration

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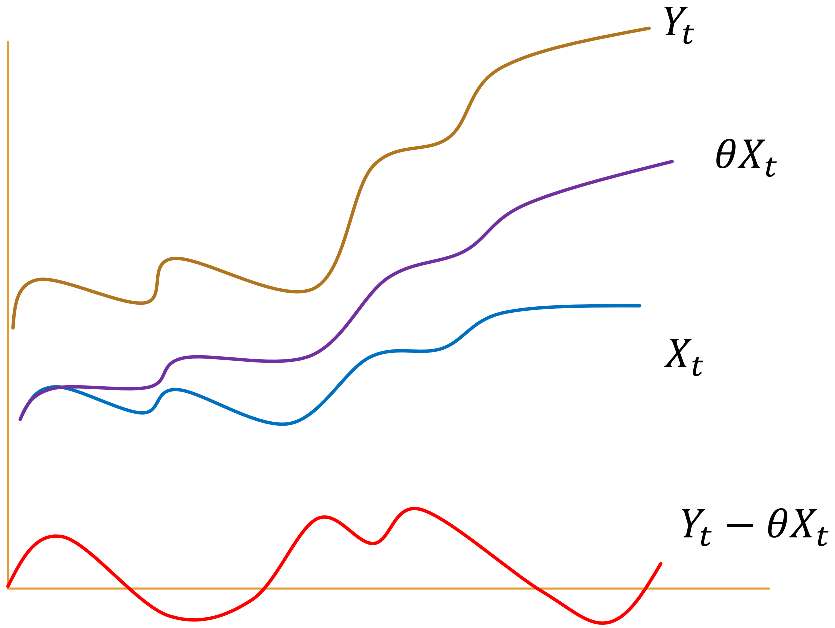
$$Y_t = a_1 + b_1 Z_t + \epsilon_1$$

$$Z_t \ is \quad \mathrm{I(1)}$$

$$X_t = a_2 + b_2 Z_t + \epsilon_2$$

$$Y_t - \frac{b_1}{b_2} X_t = \underbrace{a_1 - \frac{b_1}{b_2} a_2 + \epsilon_1 - \frac{b_1}{b_2} \epsilon_2}_{\mathrm{I(0)}}$$

$$Y_t - \theta X_t \ is \ \mathrm{I(0)}$$



Engle and Granger two step procedure

- Estimate θ and residuals by doing first stage regression between Y and X
- Perform a 2nd stage regression between first differences of Y and first difference of X along with 1st lag of residual (Error Correction Model) and test the significance of the coefficient of residual

$if\ \ Y_t - \theta X_t\ \ is$ I(0)

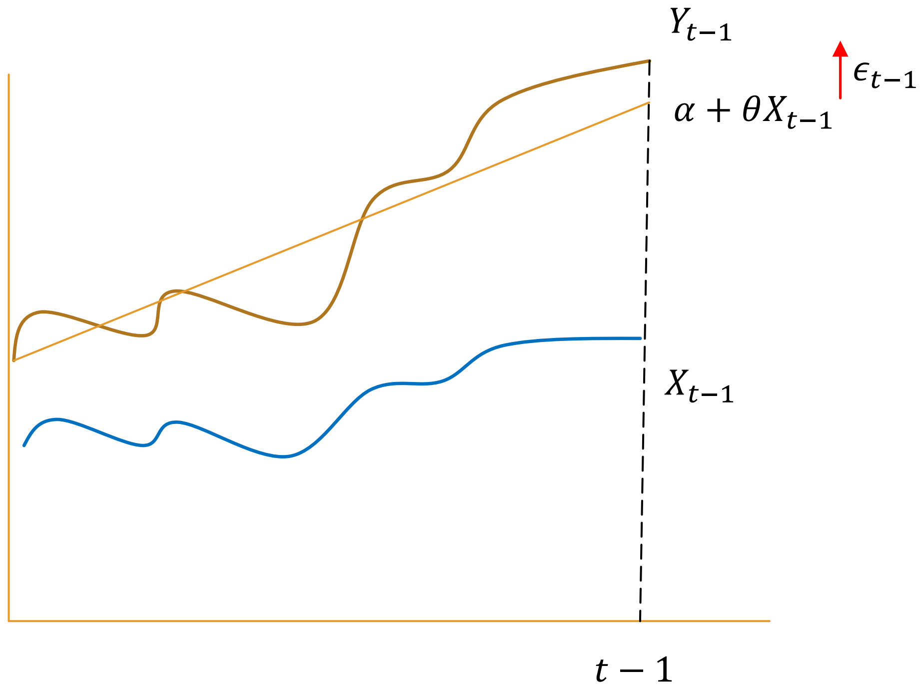
$Y_t = \alpha + \theta X_t + \epsilon$ Regress to get $\hat{\theta}$ and $\hat{\epsilon}$

$\Delta Y_t = a + b\Delta X_t + u$ short run dynamics

$$\Delta Y_t = \underbrace{a' + b'\Delta X_t}_{\text{short run dynamics}} - \lambda\underbrace{(Y_{t-1} - \alpha - \theta X_{t-1})}_{\text{long run dynamics}} + u$$

$\Delta Y_t = a' + b'\Delta X_t - \lambda(\hat{\epsilon}_{t-1}) + u$ Error correction model

$test\ if\ \lambda\ \ is\ significant$



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Thanks for Watching!